

2.4. Maintenance

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2.4.1. Maintenance Schedule

2.4.1.1. General

This section describes when and which inspection and maintenance actions have to be carried out. In case you need more information, please contact the Service department of Wärtsilä NSD Nederland B.V.

The maintenance necessary for the engine depends primarily on the operating conditions. The periods stated in this schedule are guidance values only and must not be exceeded during the guarantee period. See also the sub-suppliers manuals for additional information.

2.4.1.2. Rules for inspection and maintenance

Read the rules carefully before performing any inspection and/or maintenance work.

General

- 1** When the engine is in overhaul, make sure the remote or automatic start device and external pumps are out of operation and provided with prohibiting tags.
- 2** Provide engine parts of marks for remounting in the same position. Marks on engine parts should be copied on new parts to be installed at the same location. Every exchange should be recorded in the engine logbook mentioning the reason.
- 3** Use the MANUAL during maintenance work together with the PARTS CATALOGUE.
- 4** During all maintenance work, observe the utmost cleanliness and order.
- 5** Be aware of the risk of crankcase or camshaft case explosion! Before performing any maintenance or inspection jobs on the engine, always allow the engine to cool down sufficiently. A 10 minute cooling period will do after a normal stop.
- 6** Observe the fire precautions when maintenance jobs or cleaning on the engine will be done.
- 7** Always replace locking washers, copper rings, split pins, locking wires, self locking nuts and "O" rings during assembling. In case copper rings have to be used a second time, please take care of proper annealing.
- 8** In general never leave "O"-rings mounted on spares. Store "O"-rings in a dry, cool and dark place.

- 9** Never carry out electric welding to the engine nor use the engine as conductor for welding.

Removal

- 1** Before using the hydraulic tools read section 2.4.3.1. carefully.
- 2** Before dismantling, check all pipe systems concerned are drained and pressure released. After dismantling, cover immediately all holes for lubricating oil, fuel oil and air with tape, plugs, clean cloth or similar. Engage the turning gear. Open the indicator cocks.
- 3** If heavy parts are removed from the engine, e.g. a piston and connecting rod, the crankshaft may start turning due unbalance. For safety reasons always keep the turning gear engaged throughout the jobs.
- 4** In many cases it is advisable to record clearances before disassembling.

Mounting

- 1** Make sure all parts are carefully clean (free of carbon deposit) before mounting. Do not use cotton waste for inside cleaning of engine but use lint free cleaning rags.
- 2** Before using the hydraulic tools read section 2.4.3.1. carefully.
- 3** For advised glues, lubricants, sealants, see the parts catalogue. Never use other lubricants for bolt connections of engine components than those advised. Tightening torques will strongly differ if lubricants of different brand or type are used.
- 4** Before fitting spare parts, available as complete sub-assemblies, all "O" rings integrated must be inspected on ageing and damage, and replaced if necessary.
- 5** In general, all piping should be carefully cleaned before installing. Fuel, lubricating oil and air lines should be acid cleaned and neutralized. Also after heating of pipe sections iron oxidation has to be removed by pickling (acid cleaning). Please contact our Service Inspection Department for detailed information.
- 6** Fit all piping stress free.

Last check

- 1** It is important no tools, parts or other foreign matters have been left in or on the engine and all parts have been thoroughly cleaned before closing the engine.
- 2** Due to the compression ability of some gasket material the prestress of some flange connections should be checked after 24 hours of operation.
- 3** Record engine parts renewed.
- 4** Record engine running hours.

2.4.1.3. Maintenance schedule

2.4.1.3.1. General

The total running hours between overhauls as well as the effective life time of components depend generally on the following aspects:

- The operation and maintenance of the engine should be in accordance with instructions as specified in the engine documentation.
- Everyone concerned with the maintenance of the engine should be trained and qualified for the job and should have the engine documentation available.
- To ensure the efficiency, reliability and lifetime of the engine and its components, only genuine spare parts should be used.
- The load of the engine should be between 60% and 100% of the output, indicated on the engine type plate, making 3,000 to 6,000 running hours annually.
- Quality and treatment of lubricating oil, fuel, cooling water and air should be on accordance with the rules.

Deviation from the above factors may result in adapting the running hours between overhauls and/or effective life time of the components.

2.4.1.3.2. Procedure for inspection and maintenance

Inspection

- If, for example, two valve rotators have to be replaced inspect also two rotators of an other cylinder unit. If the result of this second inspection is also negative replace all valve rotators.
- The Job code is for internal use of the Wärtsilä NSD Nederland B.V. Service department. See example below.
- The results of the first 4.000 hours inspection are used to establish the further service intervals.

General maintenance points												
See	Description	D A I L Y	W E E K L Y	Running hours								
				S P E C I A L	5 0 0 0	1 0 0 0	2 0 0 0	4 0 0 0	8 0 0 0	1 2 0 0	2 4 0 0	3 6 0 0
	Check visually engine on leakages, bolt connections and cables/wiring.	●	●		●	●	●	●	●	●	●	●
	Check telltale hole of charge air receiver on possible water.	●	●		●	●	●	●	●	●	●	●
2.3.4.1.2.	Record performance data in engine log sheet and check for deviations.	●	●		●	●	●	●	●	●	●	●
	Check fluid levels.	●	●		●	●	●	●	●	●	●	●
2.3.4.3.	Record combustion pressures.							●	●	●	●	●
Engine arrangement												
	Check foundation bolts and chocks - first half year every 2 months. - at interval of			●								
								●	●	●	●	●
	Check reference points of epoxy resin chocks (if applicable).							●	●	●	●	●
	Inspect elastic engine mounting elements (if applicable).							●	●	●	●	●
	Elastic elements of flexible coupling: - inspect visually. - inspect at interval of: - renew.			●	= first 4,000							
									●	●	●	
				●	= 48,000							
	Measure axial displacement and crankshaft deflections of crankshaft.							●	●	●	●	●
	Check alignment between engine and engine driven machinery.							●	●	●	●	●

Fuel System												
See	Description	D A I L Y	W E E K L Y	Running hours								
				S P E C I A L								
					5 0 0 0	1 0 0 0	2 0 0 0	4 0 0 0	8 0 0 0	1 2 0 0	2 4 0 0	3 6 0 0
	Check the leak fuel quantity of the fuel system.	●	●		●	●	●	●	●	●	●	●
	Fuel filter: - check pressure difference, renew (MDO) / clean (HFO) if necessary. - renew filter elements.	●	●		●	MDO HFO						
				MDO HFO		●	●	●	●	●	●	●
	Drain impurities and condensate of fuel day tank.	●	●		●	●	●	●	●	●	●	●
	Renew O-rings of fuel supply and return lines.									●	●	●
	Inspect / overhaul fuel circulating pump.	MDO									●	
	Inspect fuel pump support: - one - all									●		●
											●	

Lubricating oil system												
See	Description	D A I L Y	W E E K L Y	Running hours								
				S P E C I A L	5 0 0 0	1 0 0 0	2 0 0 0	4 0 0 0	8 0 0 0	1 2 0 0	2 4 0 0	3 6 0 0
	Sample / analyze lubricating oil; record analyze results, lubricating oil service time consumption and charge changes.			●	= 250							
1.2.2.3.2.	Inspect / overhaul thermostatic valves: - check function - renew										●	●
1.2.2.3.1.	Inspect / overhaul lubricating oil pump(s).										●	
1.2.2.3.2.	Inspect / overhaul lubricating oil cooler.										●	
	Lubricating oil filter - check pressure difference, renew if necessary. - renew filter elements + record frequency	●	●		●	●	●	●	●	●	●	●
1.2.2.3.3.	Clean centrifugal filter. Renew all seals of centrifugal filter.		●		●	●	●	●	●	●	●	●
Starting air system												
	Drain starting air vessels and air treatment unit(s).	●	●		●	●	●	●	●	●	●	●
	Check automatic servo lubricator on oil consumption after every start.			●								
	Inspect compressed air starter motor.									●	●	●
	Inspect / overhaul compressed air starter motor and automatic servo lubricator.			●	= 60,000							

Cooling water system												
See	Description	D A I L Y	W E E K L Y	Running hours								
				S P E C I A L	5 0 0 0	1 0 0 0	2 0 0 0	4 0 0 0	8 0 0 0	1 2 0 0	2 4 0 0	3 6 0 0
	Control condition of HT and LT cooling water.		●		●	●	●	●	●	●	●	●
1.4.4.3.	Check working of cooling water venting system.									●	●	●
1.2.2.3.2.2.	Thermostatic valve: - check function - renew										●	●
	Inspect / overhaul cooling water pumps. Renew seals and bearings.										●	

Charge air and exhaust gas system											
See	Description	D A I L Y	W E E K L Y	Running hours							
				S P E C I A L	1	2	4	8	1	2	3
				5	0	0	0	0	2	4	6
				0	0	0	0	0	0	0	0
				0	0	0	0	0	0	0	0
Turbocharger											
1.5.6.3.	Clean turbine by water injection.		HFO	●	= 24 ... 200						
	Inspect and clean turbocharger.			●	= first 4,000						
	Clean compressor by water injection.	●	●		●	●	●	●			
	Inspect and overhaul turbocharger(s).							●		●	
	Inspect and clean air filter of turbocharger (if applicable).		●		●	●	●	●			
	Renew air filter material (if applicable).								●	●	●
Charge air cooler											
	Make visual inspection air side and inspect the water side.			●	= first 4,000						
	Make inspection of air side.						●	●			
	Inspect / overhaul charge air cooler.								●	●	●
Exhaust gas ducting											
	Inspect exhaust gas manifolds, expansion bellows, etc.									●	●
	Inspect exhaust gas manifolds etc. Renew expansion bellows.			●	= 48,000						

Control system												
See	Description	D A I L Y	W E E K L Y	Running hours								
				S P E C I A L	5 0 0 0	1 0 0 0	2 0 0 0	4 0 0 0	8 0 0 0	1 2 0 0	2 4 0 0	3 6 0 0
	Fuel control mechanism linkage: - check / lubricate.		●		●	●	●	●	●	●		●
	- inspect / overhaul.										●	
Governor, governor drive												
	Refresh governor lubricating oil charge. Inspect / overhaul governor and governor drive.							●	●	●	●	●
											●	
Alarm and safety system												
	Check if alarm & safety system is on with running engine and after every start.	●	●		●	●	●	●	●	●	●	●
	Check functioning of the alarm & safety system and measuring devices.				●	●	●	●	●	●	●	●
	Calibrate measuring devices.									●	●	●
	Check functioning of the oil mist detector (if applicable).	●	●		●	●	●	●	●	●	●	●

Operation											
See	Description	D A I L Y	W E E K L Y	Running hours							
				S P E C I A L	1	2	4	8	1	2	3
				5 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	2 0 0 0	4 0 0 0	6 0 0 0
	Test start process for engine in stand-by position.		●		●	●	●	●	●	●	●
	Make test run.						●	●			
	Make test run. Follow running-in program and make functional test.								●	●	●

Engine block with bearings													
See	Description	D A I L Y	W E E K L Y	Running hours									
				S P E C I A L							1	2	3
				5 0 0 0	1 0 0 0	2 0 0 0	4 0 0 0	8 0 0 0	2 0 0 0	4 0 0 0	6 0 0 0		
	Inspect crankcase visually.								●	●	●	●	
	Inspect cooling water spaces.	MDO	HFO								●	●	
2.6.2.4.	Vibration damper(s): - sample fluid.			●	= first 12,000 hereafter next interval as advised by test result.								
	- renew.			●	= 48,000								
Main bearing													
2.5.2.3.	Inspect one main bearing and main journal.	MDO	HFO								●	●	
2.5.2. + 2.5.3.	Renew main bearings and inspect main journals.	MDO	HFO								●	●	
Camshaft bearing													
2.5.4.	Inspect one camshaft bearing and journal.										●		
2.5.4.	Renew camshaft bearings and inspect journals.											●	

Crankshaft, connecting rod, piston and liner													
See	Description	DAILY	WEEKLY	Running hours									
				SPECIALLY	5000	1000	2000	4000	8000	12000	24000	48000	96000
2.6.2.5.	Inspect one: piston, (without dismantling of piston rings), connecting rod, crankpin with crankpin bearing, cylinder liner, gudgeon bearing and gudgeon.			●	=first 4,000								
Connecting rod													
	Gudgeon bearings and gudgeon: - inspect.	MDO	HFO								●	●	●
	- renew.			●	= 48,000								
	Inspect one crankpin journal with crankpin bearing.	MDO									●		●
	Inspect and overhaul connecting rods.	MDO	HFO								●	●	●
	Renew crankpin bearings and inspect crankpin journals.	MDO	HFO								●	●	●
Piston													
2.6.2.	Inspect and overhaul pistons and renew piston rings.	MDO	HFO								●	●	●
2.6.2.	Renew piston.	MDO	HFO	●	= 96,000 = 60,000								
Cylinder liner													
	Inspect and overhaul cylinder liners and renew anti bore polishing rings.	MDO	HFO								●	●	●
2.6.2.	Renew cylinder liner with anti bore polishing ring.	MDO	HFO	●	= 96,000 = 60,000								
Cranking device													
	Check operation of cranking device and start interlock device.				●	●	●	●	●	●	●	●	●
	Refresh lubricating oil charge.										●		

Cylinder head with valves												
See	Description	D A I L Y	W E E K L Y	Running hours								
				S P E C I A L								
					5 0 0 0	1 0 0 0	2 0 0 0	4 0 0 0	8 0 0 0	1 2 0 0	2 4 0 0	3 6 0 0
	Make endoscopic inspection of valves, valve seats and running surface of cylinder liners.		MDO HFO					● ●	● ●	●		●
	Inspect safety valves.										●	
Cylinder head												
2.7.2.1.	Inspect/overhaul cylinder head			●	= first 4.000							
	- one											
	- all at interval of:		MDO HFO								● ●	●
Valves												
2.7.2.4.	Check valve clearances:			●	= first 100							
	- on new and overhauled engines.											
	- and at interval of:					●	●	●	●	●	●	●
	Check function of valve rotators.						●	●	●	●	●	●
2.7.4.2.	Inlet valves:											
	- inspect / overhaul		MDO HFO								● ●	
	- renew		MDO HFO	●	= 48,000							
												●
2.7.4.2.	Exhaust valves:											
	- inspect / overhaul		HFO							●		●
	- renew										●	

Camshaft and valve drive mechanism											
See	Description	D A I L Y	W E E K L Y	Running hours							
				S P E C I A L	1	2	4	8	1	2	3
				5 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	0 0 0 0	2 0 0 0	4 0 0 0	6 0 0 0
	Inspect camshaft(s).						●	●	●	●	●
Driving gear											
	Inspect driving gear camshaft; - after			●	= first 4,000						
	- at interval of:								●	●	●
	Inspect driving gear built-on pumps; - after			●	= first 4,000						
	- at interval of:								●	●	●
Valve drive mechanism											
2.8.3.2.	Inspect inlet/exhaust roller holder; - one								●		●
	- all at interval of:									●	
	Check pushrod pivots.								●	●	●
	Inspect and overhaul rocker arms and support.									●	

Injection system												
See	Description	D A I L Y	W E E K L Y	Running hours								
				S P E C I A L	5 0 0 0	1 0 0 0	2 0 0 0	4 0 0 0	8 0 0 0	1 2 0 0	2 4 0 0	3 6 0 0
2.9.3.	Inspect, overhaul + test fuel injector.							●	●	●	●	●
	Check the fuel injection timing.									●	●	●
2.9.2.	Inspect conical sealings and cavitation on inside of the HP fuel lines.									●	●	●
HP fuel pump												
2.9.1.	- Inspect and test one.			●	= first 4,000							
	- Inspect, overhaul and test all.		MDO HFO							●	●	●

2.4.2. Maintenance Tools

2.4.2.1. General

Maintenance of a diesel engine requires a number of special tools developed in the course of engine design. Some of these tools are supplied with the engine and others are available through our service stations or for direct purchase by the customer.

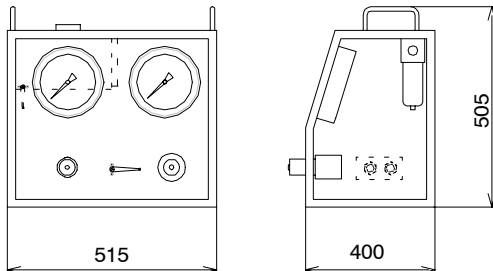
Tool requirements for a particular installation may vary greatly depending on the use and service area. Special tools are therefore selected to meet basic requirements.

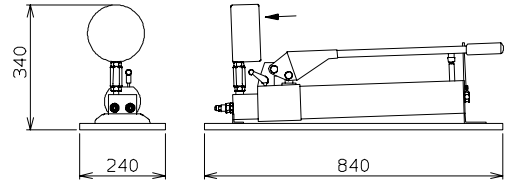
This list represents a selection of tools for the engine.

The tools are grouped in order to facilitate selection for specific service operations.

Miscellaneous tools, as mentioned in section 2.4.2.2., are used in various chapters.

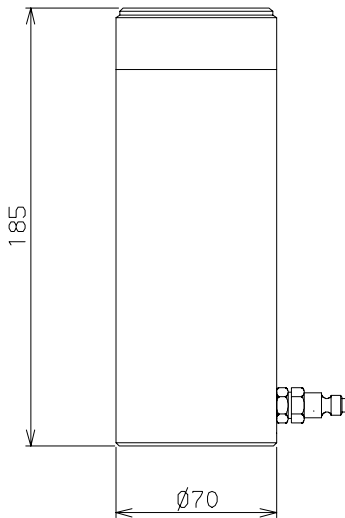
2.4.2.2. Miscellaneous tools

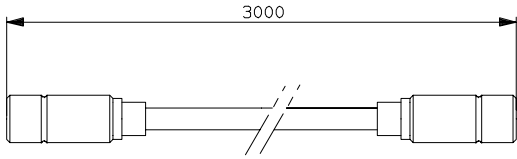
Description	Code number	Weight kg	Dimensions
Hydraulic pump HP (air operated)	9612ZT380		

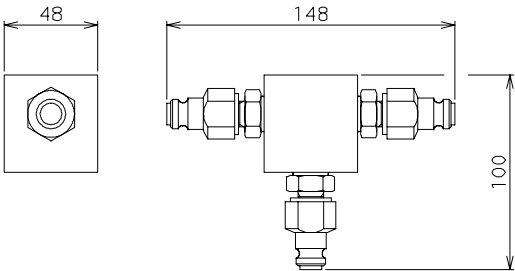
Description	Code number	Weight kg	Dimensions
Hydraulic pump HP (manually operated)	9612ZT955		

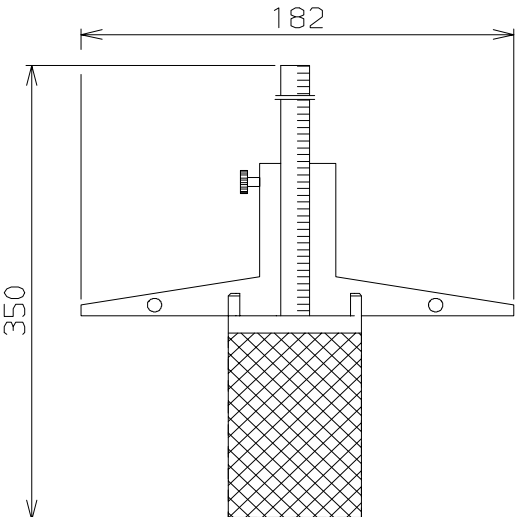
Description	Code number	Weight kg	Dimensions
Hydraulic jack for: – main bearing studs – side studs – cylinder head studs Consisting of: 1. Knurled nut 2. Tool pin	9612ZT901 9612ZT119 9612ZT125		

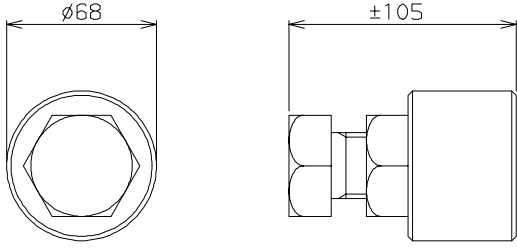
Description	Code number	Weight kg	Dimensions
Tool pin for hydraulic jacks	9612ZT125		

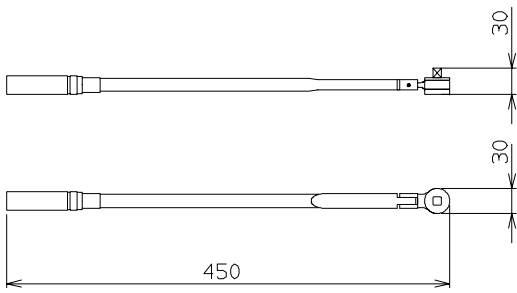
Description	Code number	Weight kg	Dimensions
Hydraulic jack In combination with: – cooler mounting equipment – camshaft bearing bush extractor – valve assembling tool	9612ZT913 9612ZT928 9612ZT914 9612ZT919		

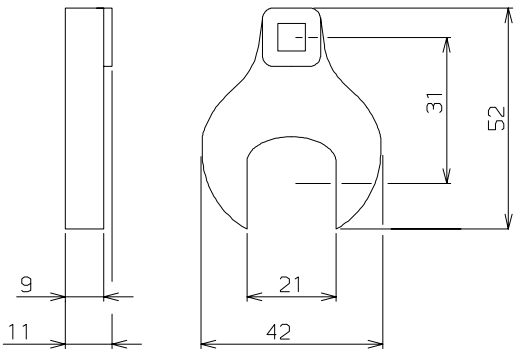
Description	Code number	Weight kg	Dimensions
Flexible hose HP	9612DT381		

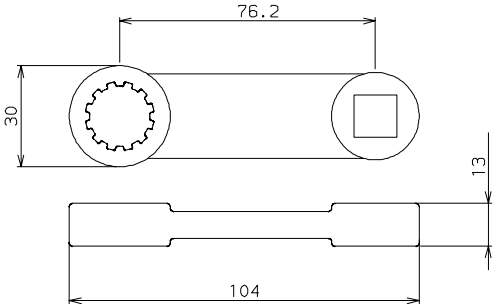
Description	Code number	Weight kg	Dimensions
Distribution block HP hoses	9612ZT918		

Description	Code number	Weight kg	Dimensions
Depth gauge for: – cylinder head stud – main bearing stud – side stud	9612ZT941		

Description	Code number	Weight kg	Dimensions
Remover main bearing and cylinder head studs	9612ZT935		

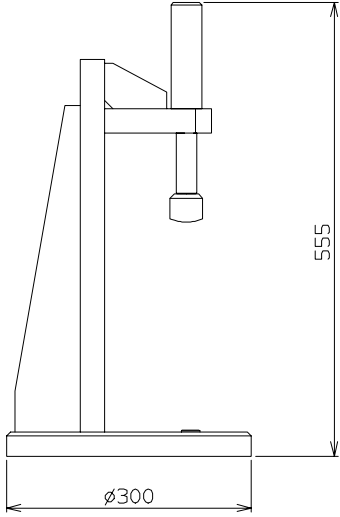
Description	Code number	Weight kg	Dimensions
Torque spanner 22 – 140 Nm for: – multiduct – HP fuel line	9612ZT330		

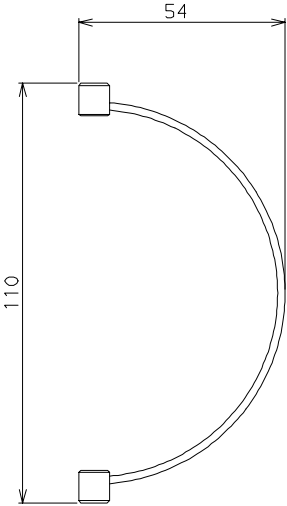
Description	Code number	Weight kg	Dimensions
Crow foot spanner for multiduct	9612ZT332		

Description	Code number	Weight kg	Dimensions
Extension for mounting cylinderhead to multiduct	9612ZT332		

2.4.2.3. Fuel system

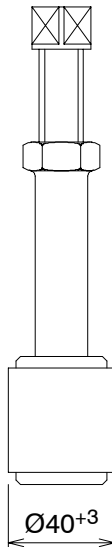
(Chapter 1.1.)

Description	Code number	Weight kg	Dimensions
Tool for fuel pump roller holder In combination with: – HP hose – hydraulic jack	9612ZT960 9612ZT382 9612ZT955		

Description	Code number	Weight kg	Dimensions
Clamp for roller holder HP fuel pump	9612ZT925		

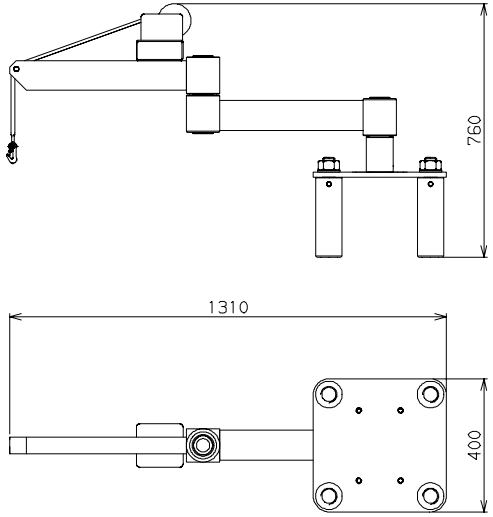
Description	Code number	Weight kg	Dimensions
Tool for mounting O-ring in support	9612ZT953		

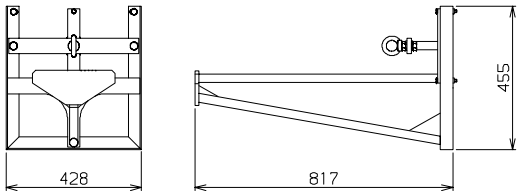
Description	Code number	Weight kg	Dimensions
Depth gauge for fuel pump push rods	9612ZT974		

Description	Code number	Weight kg	Dimensions
Tool for pulling internal pipe section	9612ZT439		

2.4.2.4. Lubricating oil system

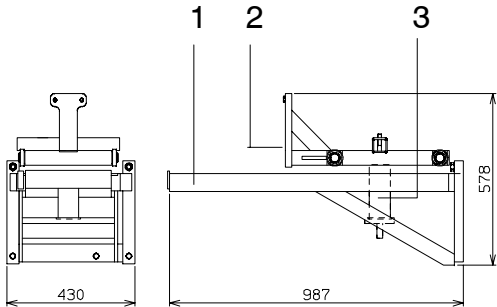
(Chapter 1.2.)

Description	Code number	Weight kg	Dimensions
Derrick for pumps engine front side	9612ZT939		

Description	Code number	Weight kg	Dimensions
Lifting extractor for lube oil cooler	9612ZT922		

2.4.2.5. Exhaust gas and charge air system

(Chapter 1.5.)

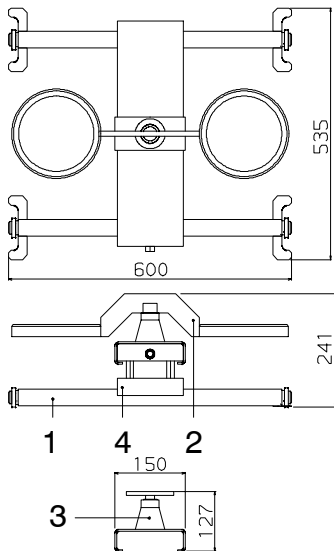
Description	Code number	Weight kg	Dimensions
Tool for removal charge air cooler consisting of: 1. support 2. trolley In combination with: 3. hydraulic jack	9612ZT928 9612ZT101 9612ZT102 9612ZT913		

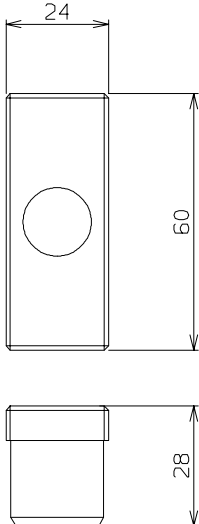
2.4.2.6. Engine block with main bearings

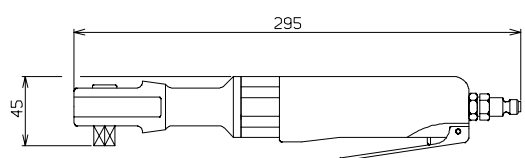
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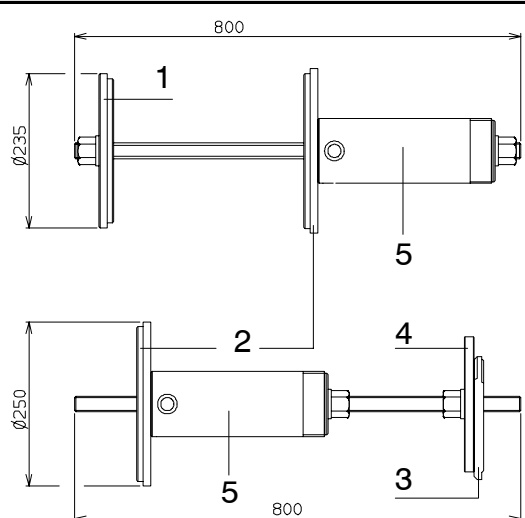
Description	Code number	Weight kg	Dimensions
Tool for side stud consisting of: 2. tie rod 4. distance piece In combination with: 1. hydraulic jack 3. knurled nut 5. tool pin	9612ZT902 9612ZT129 9612ZT130 9612ZT901 9612ZT119 9612ZT125		

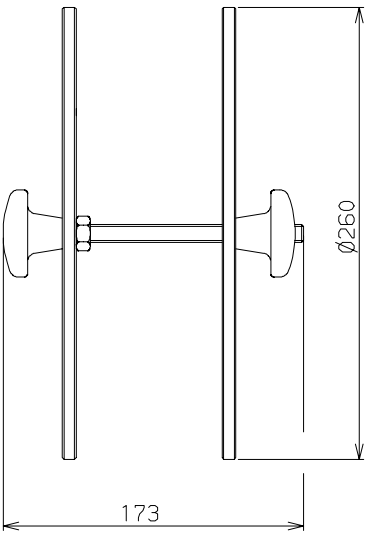
Description	Code number	Weight kg	Dimensions
Tool for main bearing consisting of: 2. tie rod 4. distance piece In combination with: 1. hydraulic jack 3. knurled nut 5. tool pin	9612ZT903 9612ZT127 9612ZT128 9612ZT901 9612ZT119 9612ZT125		

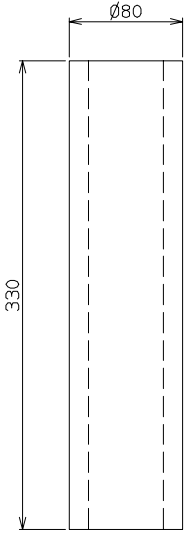
Description	Code number	Weight kg	Dimensions
Trolley for main bearing jack consisting of: 1. girder 2. top plate 3. support 4. lifting tool	9612ZT906 9612ZT261 9612ZT262 9612ZT263 9612ZT267		

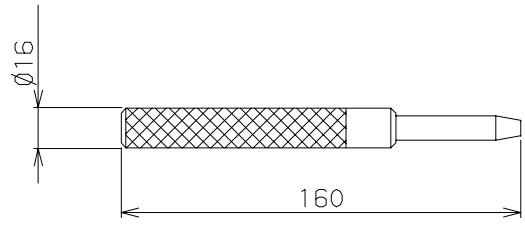
Description	Code number	Weight kg	Dimensions
Bearing shell driver	9612ZT910		

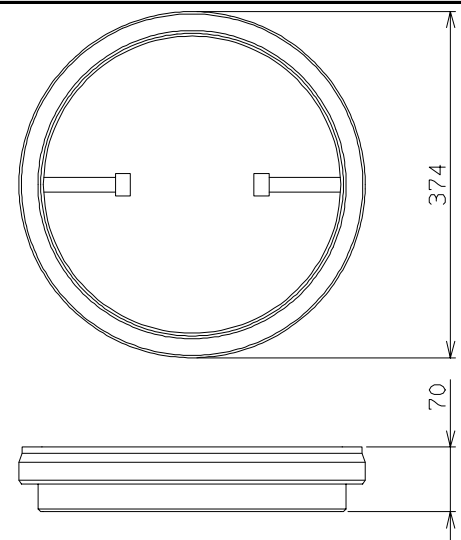
Description	Code number	Weight kg	Dimensions
Air driven ratchet for trolley main bearing	9612ZT334		

Description	Code number	Weight kg	Dimensions
Extractor tool camshaft bearing bush consisting of: 1. disc 2. support disc 3. disc axial bearing 4. support disc for axial / end bearings In combination with: 5. hydraulic jack	9612ZT914 9612ZT151 9612ZT152 9612ZT153 9612ZT154 9612ZT913		

Description	Code number	Weight kg	Dimensions
Clamp for mounting camshaft axial / end bearings	9612ZT948		

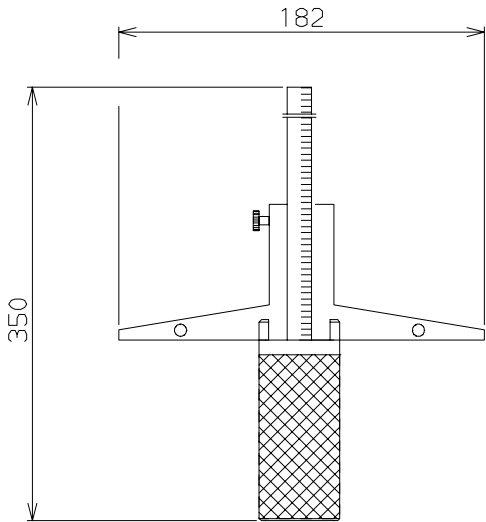
Description	Code number	Weight kg	Dimensions
Distance pipe for cylinder liner fixation	9612ZT256		

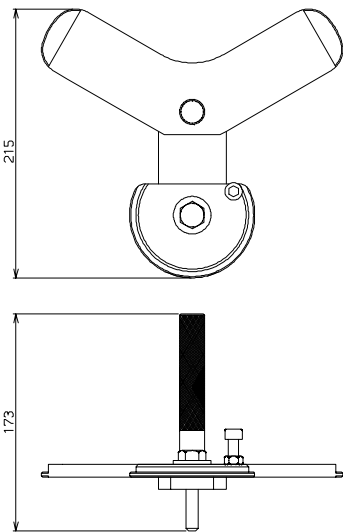
Description	Code number	Weight kg	Dimensions
Positioning pin for camshaft bearing bush	9612ZT194		

Description	Code number	Weight kg	Dimensions
Grinding ring for: – contact surface on cylinder block for liner – cylinder head gas sealing surface	9612ZT950		

2.4.2.7. Crankshaft, connecting rod, piston and liner

(Chapter 2.6.)

Description	Code number	Weight kg	Dimensions
Depth gauge for connecting rod studs	9612ZT940		

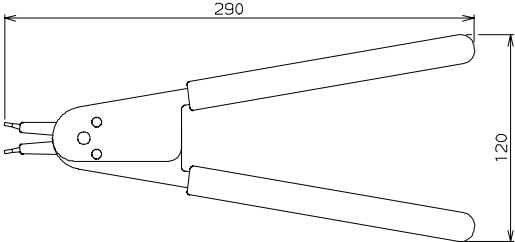
Description	Code number	Weight kg	Dimensions
Extractor anti bore polishing ring	9612ZT915		

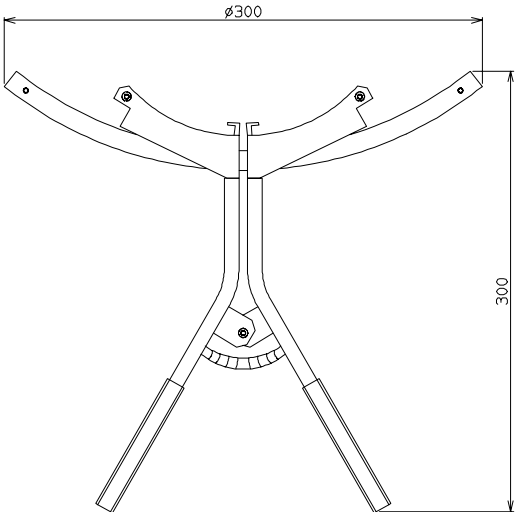
Description	Code number	Weight kg	Dimensions
Tool for big end bearing consisting of: 1. jack In combination with: 2. tie rod 3. knurled nut 4. tool pin	9612ZT944 9612ZT476 9612ZT477 9612ZT125		

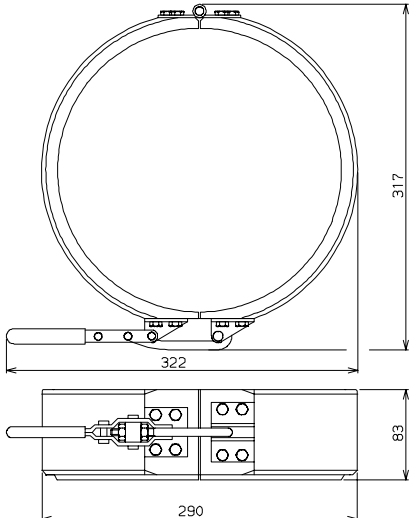
Description	Code number	Weight kg	Dimensions
Lifting tool for connecting rod/piston/liner assembly In combination with: 1. protecting sleeves 2. positioning bracket 3. protecting sleeves	9612ZT911 9612ZT114 9612ZT113 9612ZT112		

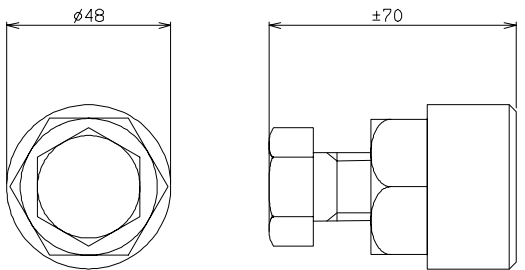
Description	Code number	Weight kg	Dimensions
Extractor tool cylinder liner consisting of: 1. tie rod 2. ring 3. nut 4. bracket In combination with: – lifting tool	9612ZT912 9612ZT160 350 007 9612ZT156 9612ZT159 9612ZT911		

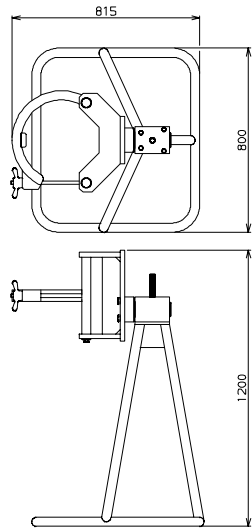
Description	Code number	Weight kg	Dimensions
Tap M8 threaded holes in piston crown	9612ZT195		

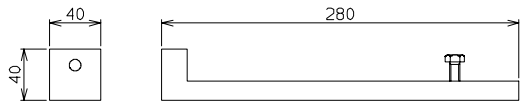
Description	Code number	Weight kg	Dimensions
Circlip plier piston	9612ZT250		

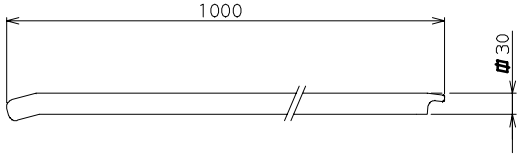
Description	Code number	Weight kg	Dimensions
Piston ring plier	9612ZT185		

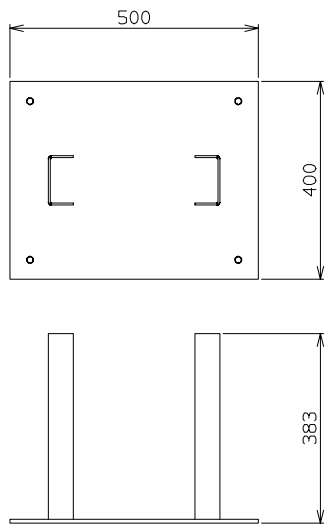
Description	Code number	Weight kg	Dimensions
Piston ring mounting tool	9612ZT917		

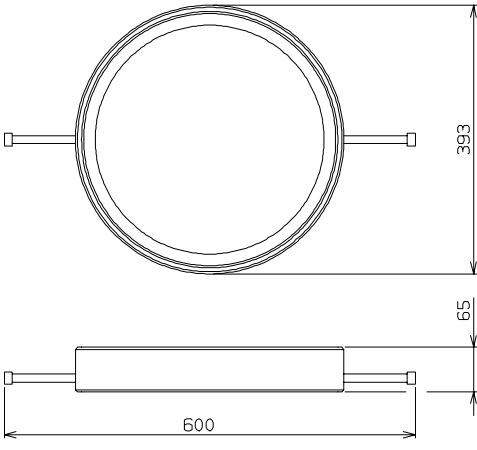
Description	Code number	Weight kg	Dimensions
Remover connecting rod studs	9612ZT936		

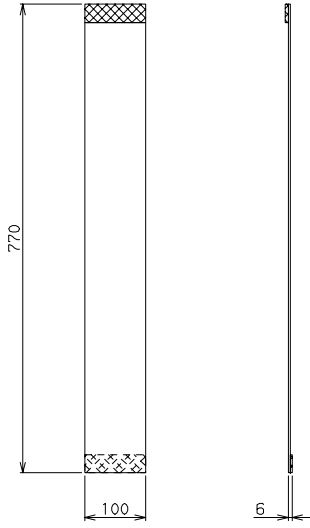
Description	Code number	Weight kg	Dimensions
Tool for connecting rod/piston/liner assembly	9612ZT954		

Description	Code number	Weight kg	Dimensions
Piston support	9612ZT938		

Description	Code number	Weight kg	Dimensions
Crow bar for axial clearance crankshaft	9612ZT376		

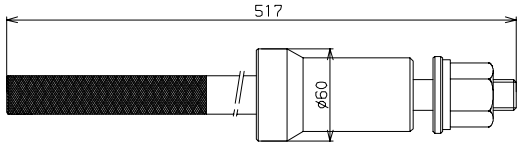
Description	Code number	Weight kg	Dimensions
Support for connecting rod / piston assembly	9612ZT445		

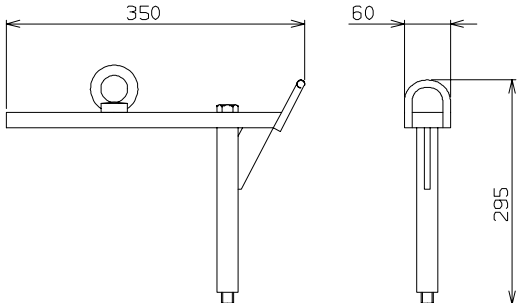
Description	Code number	Weight kg	Dimensions
Grinding ring cylinder liner top	9612ZT951		

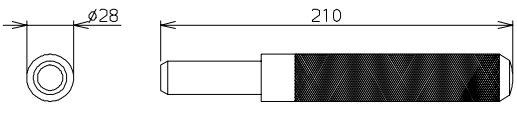
Description	Code number	Weight kg	Dimensions
Crankshaft protecting canvas	9612ZT970		

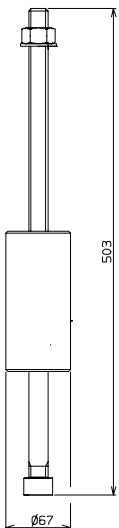
2.4.2.8. Cylinder head with valves

(Chapter 2.7.)

Description	Code number	Weight kg	Dimensions
Mounting tool injector sleeve	9612ZT946		

Description	Code number	Weight kg	Dimensions
Lifting bracket for cylinder head	9612ZT927		

Description	Code number	Weight kg	Dimensions
Tool fitting cooled valve guides	9612ZT233		

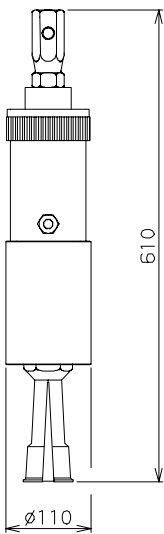
Description	Code number	Weight kg	Dimensions
Extracting tool valve guide	9612ZT909		

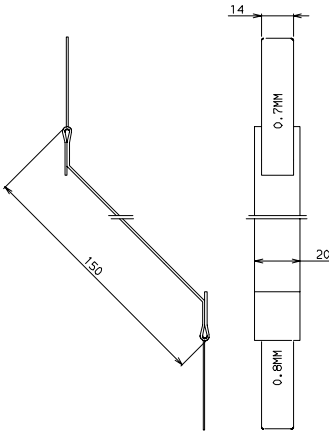
Description	Code number	Weight kg	Dimensions
Spanner for removal valve rotator	9612ZT956		

Description	Code number	Weight kg	Dimensions
Mounting tool valve seat	9612ZT945		

Description	Code number	Weight kg	Dimensions
Tool for cylinder head stud consisting of: 2. tie rod 4. distance piece In combination with: 1. hydraulic jack 3. knurled nut 5. tool pin	9612ZT904 9612ZT131 9612ZT132 9612ZT901 9612ZT119 9612ZT125		

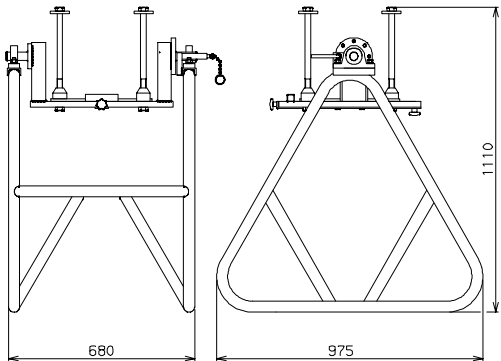
Description	Code number	Weight kg	Dimensions
Spanner indicator cock	9612SW510		

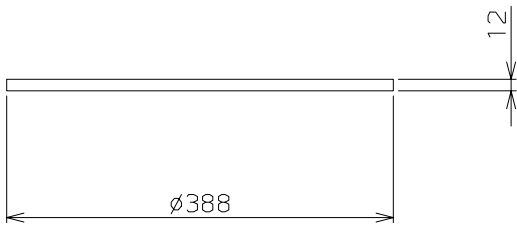
Description	Code number	Weight kg	Dimensions
Extractor injector sleeve	9612ZT932		

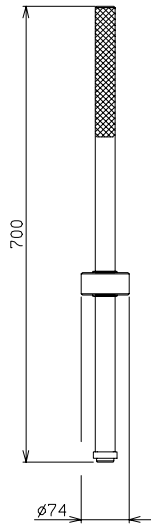
Description	Code number	Weight kg	Dimensions
Valve clearance feeler	9612ZT251		

Description	Code number	Weight kg	Dimensions
Valve assembling tool consisting of: – spindle – pressing plate HFO – pressing plate MDO In combination with: – jack	9612ZT919 9612ZT147 9612ZT184 9612ZT145 9612ZT913		

Description	Code number	Weight kg	Dimensions
Lifting frame cylinder head jacks	9612ZT943		

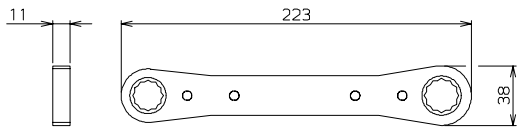
Description	Code number	Weight kg	Dimensions
Cantling frame	9612ZT958		

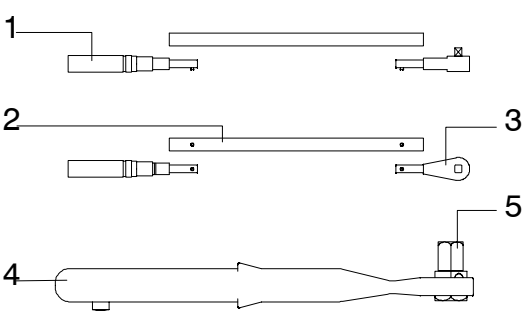
Description	Code number	Weight kg	Dimensions
Cylinder liner cover (wood)	9612ZT247		

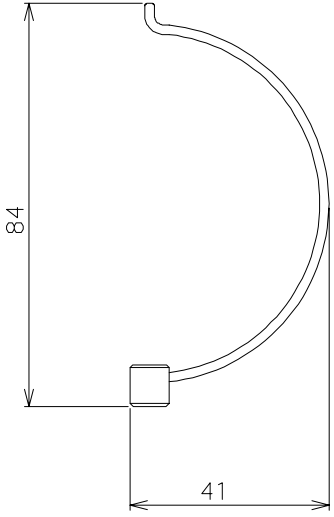
Description	Code number	Weight kg	Dimensions
Grinding tool for seat injector in cylinderhead	9612ZT962		

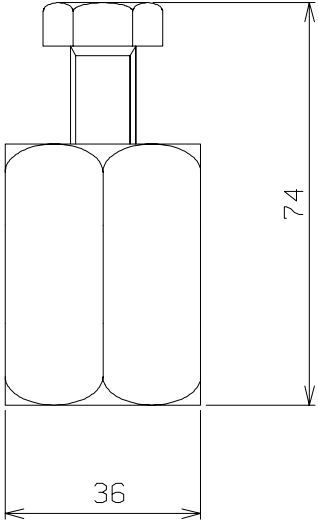
2.4.2.9. Camshaft and valve drive mechanism

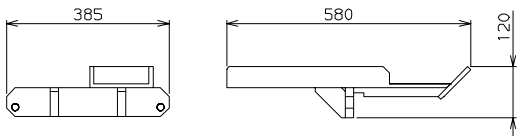
(Chapter 2.8.)

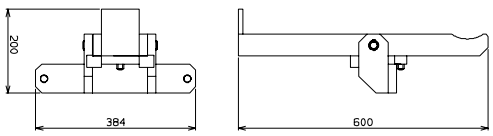
Description	Code number	Weight kg	Dimensions
Ratchet spanner for camshaft section bolts	9612ZT232		

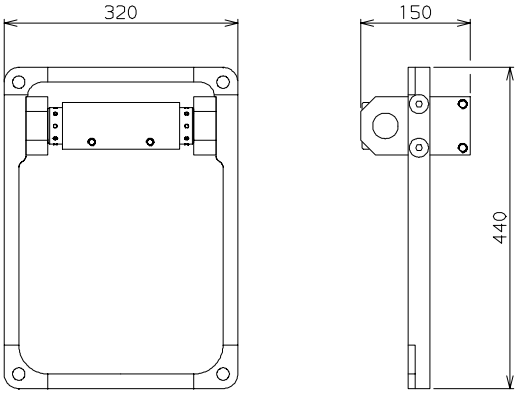
Description	Code number	Weight kg	Dimensions
Torque spanner for camshaft bolts consisting of: 1. torque spanner 2. extension 3. ratchet 4. ring spanner 5. bit Remark: Items 1, 2, 4 and 5 to be used as set.	9612ZT230		

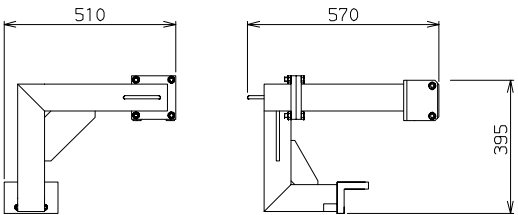
Description	Code number	Weight kg	Dimensions
Locking clamp valve roller holders	9612ZT924		

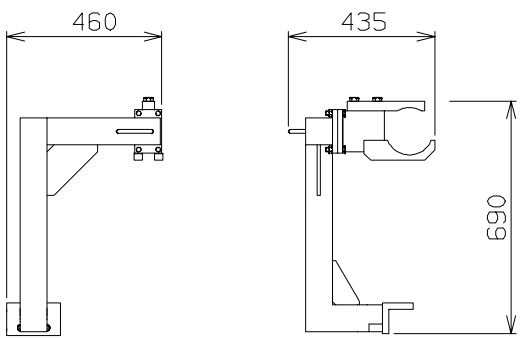
Description	Code number	Weight kg	Dimensions
Tool for disconnecting camshaft section	9612ZT388		

Description	Code number	Weight kg	Dimensions
Support for mounting camshaft journal	9612ZT957		

Description	Code number	Weight kg	Dimensions
Support for camshaft section.	9612ZT933		

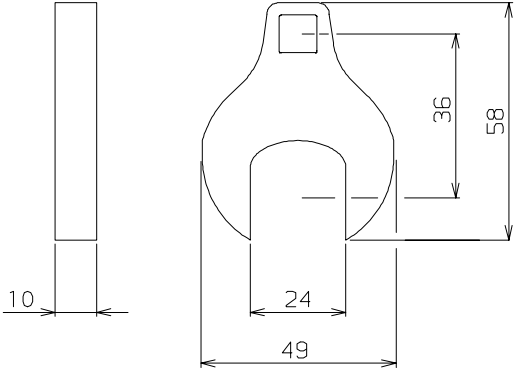
Description	Code number	Weight kg	Dimensions
Frame for tools 9612ZT965 and 9612ZT967	9612ZT963		

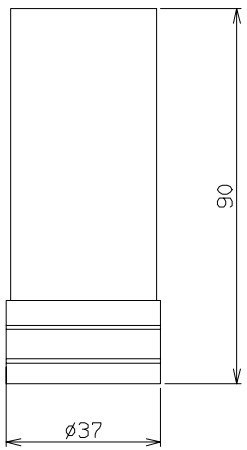
Description	Code number	Weight kg	Dimensions
Tool for removal intermediate gear wheel assembly	9612ZT965		

Description	Code number	Weight kg	Dimensions
Tool for removal camshaft gear wheel assembly	9612ZT967		

2.4.2.10. Injection system

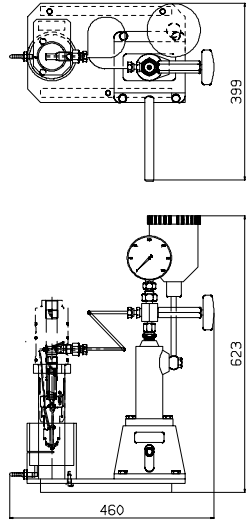
(Chapter 2.9.)

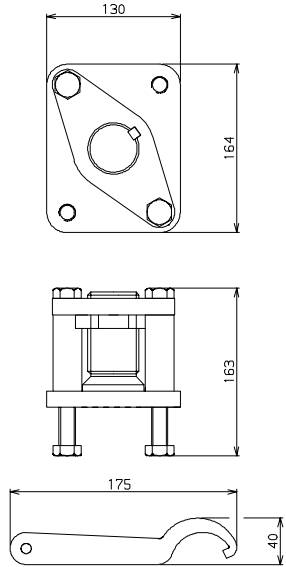
Description	Code number	Weight kg	Dimensions
Crow foot spanner for HP fuel line	9612ZT333		

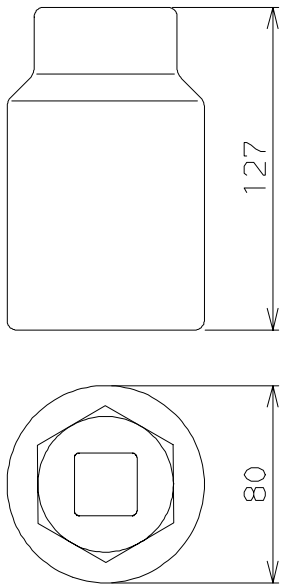
Description	Code number	Weight kg	Dimensions
Socket spanner for HP fuel line	9612ZT255		

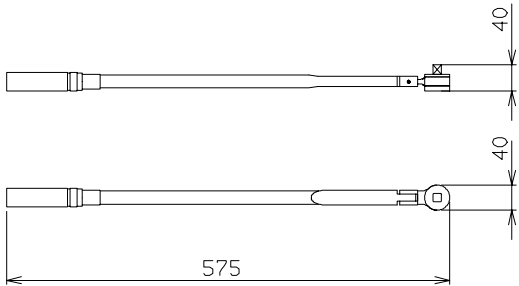
Description	Code number	Weight kg	Dimensions
Crow foot spanner HP fuel line	9612ZT331		

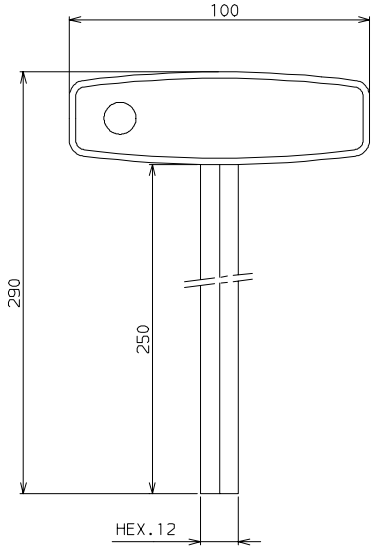
Description	Code number	Weight kg	Dimensions
Tool for injector extracting 1. bracket for manual lifting	9612ZT931 9612ZT178		

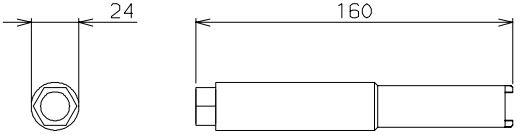
Description	Code number	Weight kg	Dimensions
Testing device injector	9612ZT976		

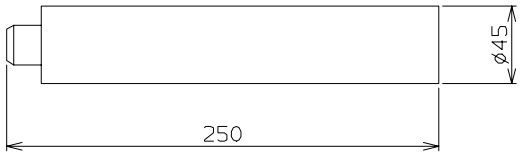
Description	Code number	Weight kg	Dimensions
Tool for HP fuel pump	9612ZT971		

Description	Code number	Weight kg	Dimensions
Socket spanner for injector sleeve nut In combination with torque spanner	9612DT259 9612ZT231		

Description	Code number	Weight kg	Dimensions
Torque spanner for injector sleeve nut, 50 – 340 Nm In combination with socket spanner	9612ZT231 9612DT255		

Description	Code number	Weight kg	Dimensions
Spanner for plug injector	9612ZT187		

Description	Code number	Weight kg	Dimensions
Adjusting tool injector	9612ZT186		

Description	Code number	Weight kg	Dimensions
Synthetic driver for HP fuel pump barrel	9612ZT225		

2.4.3. Background information of hydraulic tools and torque spanners

2.4.3.1. General

A number of important engine components are fitted by means of hydraulic tools. With the use of hydraulic tools it is possible to stress a bolt or stud of a connection till very high tension. This tightening process can be done within very small tension limits and a great force and accuracy.

It is necessary to understand the working principle of the hydraulic tools. For this reason an explanation of the principle is given.

For safety reasons it is important the tools are used in the correct way. The tightening procedures have to be carried out strictly in accordance with the instructions in this manual.

For the following subjects general background information is described:

"Rolled thread"

The thread of the studs of important connections is manufactured by a "cold rolled" process to make the stud thread strong and good resistant against fatigue. Due to this process the circumference of the thread becomes smooth and hard. This process makes studs also sensitive to breakage in case of damage. Therefore, studs must always carefully be handled. Replace a stud when damaged.

The "easy going" nut

The stud becomes a little bit longer over its entire length during elongation. This includes that also the pitch of the thread is a little extended. However, the pitch of the nut mounted is not extended. To make it possible to turn the nut on the elongated stud "quite some clearance" must exist between the thread of the nut and the thread of the stud. This clearance can be felt when turning on the nut. For this reason it must be possible, without any restriction, to turn on a nut by hand.

In case of "heavy points" remove the nut and find and solve the problem before starting the tightening procedure.

Studs

All hydraulically stretched studs are made of high tensile strength steel. To obtain the correct force in the studs, the studs have to be stretched to approximately 90% of the yield point of the material.

This means, a force 10% more than the tightening force does overstretch the stud. This may never happen because over-stretched studs become sensitive to fatigue and may break after some time.

Therefore, overstretched studs should be replaced. This is particularly important for connecting rod studs and counter weight studs.

Use genuine parts

There is a real danger in employing studs not delivered by Wärtsilä NSD Nederland B.V. in particular studs hydraulically stretched.

If these studs are not made from the same high tensile strength material and not provided with "cold rolled thread", problems may occur: Studs, made of material not sufficiently strong, will continue to elongate even if the correct oil pressure is applied. The nuts can be turned further until breaking.

Use of locking fluids

Clean parts carefully with a degreasing agent and dry parts before applying locking fluid.

Note! See for specification relevant drawing / parts list (Parts Catalogue).

The hydraulic tool set consists of :

- Hydraulic pump pneumatically driven or a manually operated pump.
- A number of hydraulic jacks, distance pieces, tie rods, knurled nuts, high pressure hoses with quick release coupling and tool pins to tighten or to loosen the nuts of the relevant components.

2.4.3.2. Pneumatic driven hydraulic pump unit

The pump unit (fig. 2.4 – 1) consists of a number of main components:

- A connection (7) for the supply of service air (maximum pressure 7 bar).
- A reducing valve (6) to adjust the service air between 0 and 6 bar.
- A manometer (5) directly mounted after the service air reducing valve.
- An air valve (4) to control the pump speed.
- An air-operated piston (3) (large surface) connected to a piston in the lower section (small surface) for pressurizing the hydraulic oil.
- A container (1) for hydraulic oil (circa 5 litres).
- A double pointer manometer (8) indicating the hydraulic jack pressure.
- A quick-release couplings with non return valve (10).
- A valve (9) in the HP oil return line.

The manometer (5) is connected to the air pressure line after the reducing valve (6). The manometer scale indicates a comparable oil pressure from 0 to 1000 bar. 6 Bar service air pressure is equivalent to 1000 bar "oil pressure" on the manometer scale. The manometer (5) makes it possible to set the required HP oil pressure before pumping.

To prevent, due to inaccuracy of this manometer, that the final obtained oil pressure will end higher than the required pressure, manometer (5) indicates 10% higher.

A manometer (8) indicates the pressure in the HP hydraulic system. This calibrated manometer contains two independently working measuring mechanisms, each provided with a scale and pointer. Recalibrating of this manometer is necessary when the indication of both pointers differs more than 10 bar.

Check the double pointer pressure gauge (8) regularly. For this purpose a calibrated pressure gauge can be delivered. This pressure gauge can be connected at the end of the HP hose.

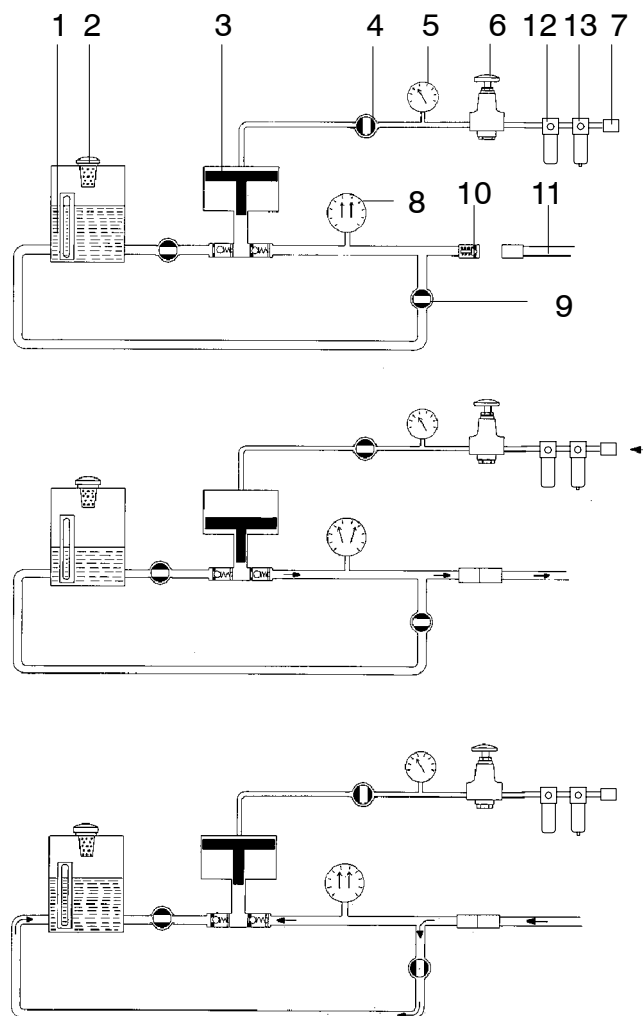


Fig. 2.4 – 1 Diagram pneumatically driven hydraulic pump unit

Operating

Always keep the pump unit horizontally. Maintain sufficiently hydraulic oil in the container to avoid air in the system and check if valve (14) is open.

During filling of the HP pump container, it is recommended to connect the set according to Fig. 2.4 – 2 . Before filling, open the release valve (9) and empty the jack (12) by pressing piston and cylinder simultaneously. Next, the container can be filled through the filling plug (2). (For specification of the hydraulic oil, see chapter 1.2.).

- 1** Close valves (4) and (9), see fig 2.4 – 1 and fig. 2.4 – 2 .
- 2** Connect at (10) the HP hose (11) to the hydraulic jacks (12).
- 3** Connect the service air at (7).
- 4** After adjusting the "air pressure" at the value of the required oil pressure, start the pump by opening air valve (4). The hydraulic system will be pressurized, visible on manometer (8). At a value of approx. 10% below the adjusted value, the pump will stop.
- 5** Slowly turn the spindle of the reducing valve (6) counter clockwise, by which the hydraulic oil pressure will increase.
- 6** Continue pumping until the reading on the manometer of the hydraulic system (8) shows the required oil pressure. The manometer in the air system will now show a pressure of circa 10 % higher.
- 7** After the correct oil pressure has been reached, push down the locking ring over the reducing valve spindle to fix the setting of the reducing valve.

Every time the pump is operated, the pressure will rise automatically and accurately till the fixed pump setting (accuracy ± 5 bar). When the hydraulic stretching procedure is completed close valve (4) and always open slowly valve (9). Quick opening of valve (4) may damage the hydraulic pressure manometer.

Remark

When the required pressure is reached and the pump continues slowly working, the hydraulic system is leaking, e.g. a leaking coupling or a leaking hydraulic jack.

When the hydraulic system contains air, disconnect the coupling at (10). Press the small valve in the coupling by means of a pin and run the pump slowly. Let oil escape as long as it contains air.

Warning!

Always check the correct connecting of hose couplings to pump unit and jacks special when more jacks are connected at the time. An incorrectly made connection may cause the jack is not connected and the jack in question not pressurized although the manometer indicates the correct pressure.

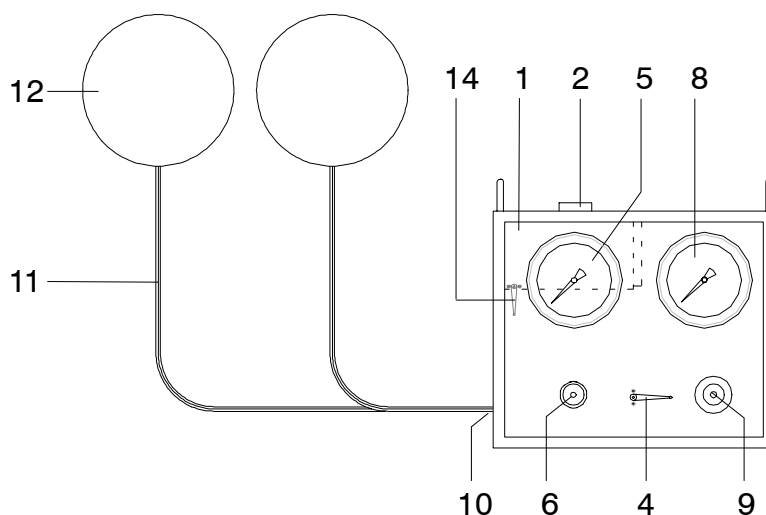


Fig. 2.4 – 2 Pneumatic driven hydraulic pump unit

2.4.3.3. Hydraulic jack

A hydraulic jack, see fig. 2.4 – 3 for an example, consists basically of a housing (3) and a piston (1) which is accommodated in a recess in the top of the housing. The piston is sealed at the top and the bottom with O-rings, (7) and (8).

The space between the O – rings can be filled with oil under pressure via connection (13). The enclosed area of the piston multiplied by the applied oil pressure results in the stretching force in the pulling piece and stud. The stud shall, as a consequence of the applied force, be lengthened. Every time when the same force (oil pressure) is applied, the same lengthening of the stud will be realised. To exert the required force in the stud, only a limited lengthening of the stud is required, a few mm only. The stroke of the jack shown is 9 mm, which can differ with the type of jack, and is sufficient for the stud elongation.

After the nut is fastened or loosened and the oil pressure is released, the stretching process can be repeated.

After the oil pressure is released it takes some time, for the springs (12) on top of the piston, to force the piston back into bottom position as friction exists in the returning of the hydraulic oil to the sump of the hydraulic pump in the small oil drillings in the jack, in the valves in the quick release couplings and the relative large quantity of oil to return to the pump.

If no time is given to displace the oil and to lower the piston till bottom position, for instance by too quickly disconnecting the hoses from the jack, the piston will stop a few mm from bottom. If this method of stretching is repeated the piston finally has no working stroke left and in the next stroke a situation arises in which the retaining cover (4) is breaking the M6 bolts (11). The retaining cover is not kept in place and will be stopped by two bolts (10).

To avoid this situation the returning of the piston into bottom position can be realized by waiting, or by using a spanner on top of the jack when mounted on a stud.

The piston is in bottom position when the top face of the jack is level with the top face of the retaining ring.

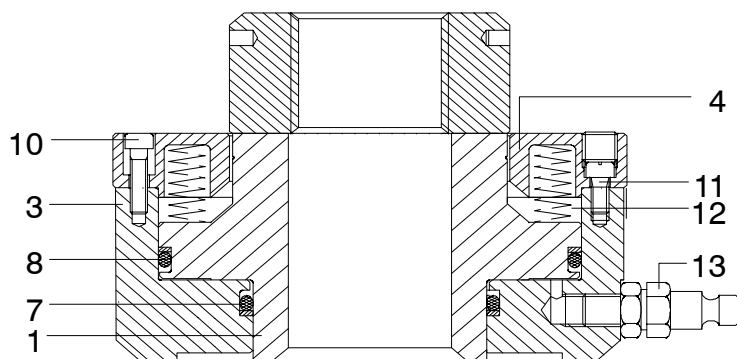


Fig. 2.4 – 3 Cross section hydraulic jack

2.4.3.4. Hydraulic extractor

For some power demanding operations a hydraulic extractor jack is applied. The jack is utilized in connection with the manual operated pump, see fig. 2.4 – 4 .

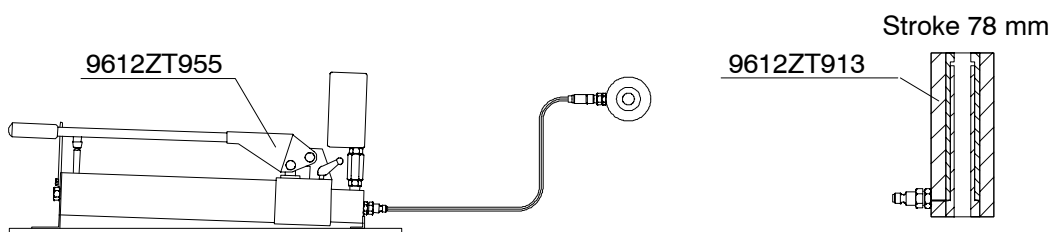


Fig. 2.4 – 4 Hydraulic jack

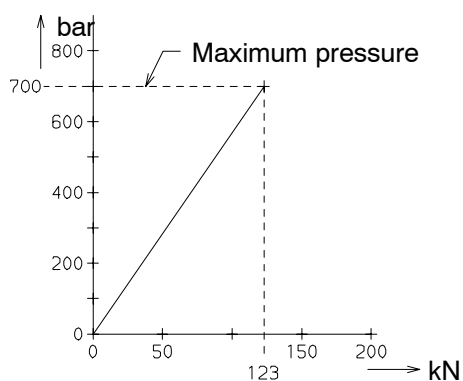


Fig. 2.4 – 5 Graphic relation pressure / force

2.4.3.5. H.P. hose

The maximum permissible hose pressure is 1500 bar.

The H.P. hose is made up of a plastic inner hose wrapped with 4 layers of steel wire, around which is a flexible, blue-coloured “skin”. This “skin” serves to make any damage to the H.P. hose more easily visible and to retain the oil stream if an internal leak develops. In this case a “blister” is formed on the hose.

Work safely!

- 1** Check the hose and quick-release couplings for damage before use.
- 2** Ensure the quick-release couplings are thoroughly clean before use. Dirt can cause damage to the quick-release couplings so leaks may develop.
- 3** After use, seal the quick-release couplings again with the dust caps (1).

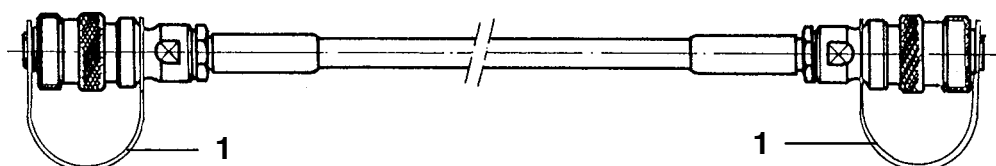


Fig. 2.4 – 6 H.P. hose

Never work with damaged hoses! The hoses will remain in good condition if you:

- 4** **never** try to remove the clamp fitting from the hose
- 5** **never** bend the hose into a radius smaller than 160 mm
- 6** **never** twist the hose
- 7** **never** damage the hose by, for example, placing heavy objects on it
- 8** **never** apply a tensile load to the hose, for example, by tugging
- 9** **never** use any oil other than is specified
- 10** **never** use the hose for other purposes.

Note! Damaged hoses or quick-release couplings should be discarded.
DO NOT CARRY OUT REPAIRS!

2.4.3.6. H.P. quick-release coupling

A complete quick-release couplings consists of:

1. male connector
2. female connector
3. spring controlled valves

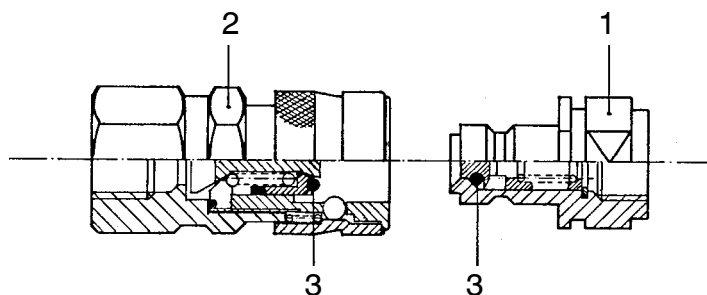


Fig. 2.4 – 7 H.P. quick-release coupling

The H.P. quick-release couplings are used to facilitate easy connecting and disconnecting of various parts of the hydraulic tool and are self-closing. This means that it is impossible for air to enter the hose or the jacks, and oil losses are minimal.

In order to guarantee the valves function efficiently, the couplings should be thoroughly clean before use. For this reason the quick-release couplings should always be sealed with dust caps (4) and (5).

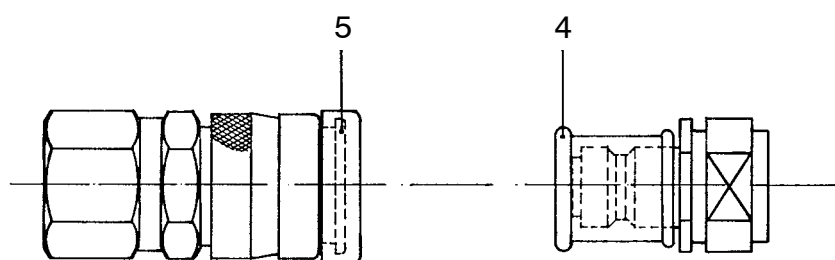


Fig. 2.4 – 8 Dust caps H.P. quick-release coupling

The couplings have a conical thread (NPT) which is self-sealing when the coupling is mounted to the hose fitting or a jack. Do not use sealing tape. Pieces of tape in the hydraulic system will impair efficient operation.

2.4.3.7. Hand-operated hydraulic pump

The hydraulic pump consists of the following main components.

1. oil container
2. pump element
3. oil pressure manometer (double indicator)
4. H.P. oil connection (2x)
5. release valve
6. filling plug
7. venting point
8. pump handle
9. locking lever for pump handle

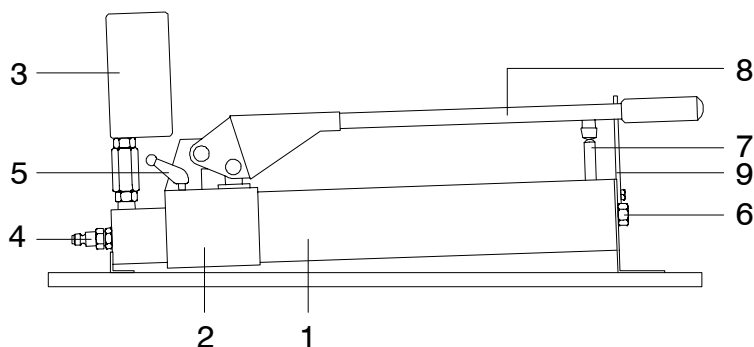


Fig. 2.4 – 9 Hand operated hydraulic pump

Operating

1 Before using check if there is sufficient hydraulic oil in the container (1) to avoid air in the system. (Always use oxidation resistant oil see chapter 1.2.1.2.)

2 Loosen locking lever (9) to operate pump handle and to open the automatic venting point (7).

3 Connect the pump to the jacks, by the H.P. hoses.

Note! **Never apply pressure to a jack which is not in use for stretching studs, otherwise the plunger will exceed its maximum stroke.**

4 Close the release valve of the pump and apply the prescribed pressure. If the pressure drops, there might be a leakage in a connection or somewhere else. The leakage should be found and repaired.

Note! **Calibrate the double indicator manometer if the difference between the indicators is more than 10 bar .**

5 Always reduce the pressure slowly to zero in order to avoid forming of foam on the oil and/or to protect the indicator.

6 During transport of the hydraulic pump lock pump handle (8) with lever (9) to close the venting point (7).

2.4.3.8. Safety instructions for hydraulic tools

The following general instructions and guides will be helpful to determine if your system components are properly connected.

- 1** Be sure all hydraulic hoses and fittings are connected to the correct inlet and outlet ports of the pump, cylinders, valves and other system components. An incorrectly made connection may cause a not connected jack in question and not pressurised although the manometer indicates the correct pressure.
- 2** Sealing. Be sure all connections are fully tightened. Seal all pipe connections with a high-grade pipe thread sealer.
- 3** Over tightening. Do not over-tighten any connection. All connections should be snug and leak-free. Excessive tightening will cause strain on threads and castings which could cause fitting failure at pressures below rated capacity.
- 4** Fully tighten hydraulic connectors (avoid excessive force). Loose connectors will act as a partial or complete line restriction causing little or no oil flow and resulting in equipment damage or failure.
- 5** If the maximum stroke is exceeded the sealing ring of the piston will open a drain hole. Pressurized oil will escape and the pressure drops to zero. If this happens the jack has to be disassembled, the sealings inspected and if damaged, replaced by new.

After assembling the jack, trapped air must be removed. Connect the jack by a flexible hose to the pump open the vent screw on the jack and place the jack with the vent opening pointing upwards to allow the air to escape. Slowly fill the jack and continue until no air bubbles escape. Close the vent screw.

- 6** Do not drop heavy objects on the H.P. hose. A sharp impact may cause bends or breaks to internal hose wire strands. Applying pressure to the damaged hose will cause internal flexing which will eventually break the hose strands causing the hose to rupture.
- 7** Do not use the hydraulic hose to carry a hydraulic component (i.e. pumps, cylinders and valves).
- 8** Avoid sharp bends and kinks when routing hydraulic hoses. If pressure is applied to a bend or kinked hose, the oil flow will be restricted causing severe back-pressure. Also the sharp bend and kinks will internally damage the hose leading to premature failure.
- 9** Avoid situations where loads are not directly centered on the cylinder plunger. Off-center loads produce considerable strain on cylinder plungers and may slip or fail causing potentially dangerous results. Avoid point loading. Distribute the load evenly across the entire saddle surface.

- 10** Always provide clearance for hoses and connectors to avoid moving objects, abrasion or sharp objects.
- 11** Use hydraulic gauges which indicate safe operating loads in the hydraulic system. Do not exceed the safe limit of the lowest rated component used within your system.
- 12** Keep hydraulic equipment away from flames and heat. Excessive heat (above 70°C) will soften packings and seals, resulting in fluid leaks.
- 13** Never attempt to lift a load which exceeds the capacity of a cylinder or jack. Overloading causes equipment failure and possible personal injury.
- 14** Do not overextend the hydraulic jack. The cylinder will take the load on the plunger stop ring. However, using the full stroke does not supply power and only adds unnecessary strain to the cylinder.
- 15** Keep oil lines clean. When connector halves are disconnected, always put on dust caps. Use every precaution to guard unit against entrance of dirt because dirt and foreign matters may cause pump, cylinder and valve failure.
- 16** Make sure no iron parts enters the engine.

2.4.3.9. Loosening of hydraulically tightened connection

Note! Loosening pressure = tightening pressure.

To loosen a connection the same oil pressure, or somewhat lower than the final pressure with which the connection has been tightened, is required.

Prevent overstretching

To prevent overstretching, the hydraulic pressure applied to the studs should never exceed the tightening pressure plus 5%. Furthermore, operate the pneumatic driven hydraulic pump unit according the instructions.

There is no reason to apply a higher pressure than the tightening pressure. When studs have been tightened to setting value, the prestress in the studs may decrease slightly during running of the engine as a result of the forces incurred in the engine components. Therefore, the oil pressure in the hydraulic jacks, required for disassembly will be the same or somewhat lower than the tightening pressure.

If one or more nuts can't be loosened at setting value, increase the oil pressure by maximum 5% and check with a feeler gauge of 0.05 mm. if the nut is still free from the contact surface. If the nut is free and it is still not possible to turn the nut, the nut is sticking due to dirt, corrosion or damage. In such case try to turn off the nut applying a hammer and a copper pin.

If still not successful don't increase the oil pressure because it serves no purpose. Applying a too high oil pressure introduces the risk of damaging the thread in the engine component in which the stud is mounted.

Therefore, remove tools, cut the nut off and remove the stud. Fit a new stud and nut.

Make sure no iron parts enters the engine.

If the nut is not free

- 1** Check if there is oil supply to all jacks.
- 2** Check the quick-release coupling and the non return valves.
- 3** Check that the tools do not touch the nuts.

If the nuts can still not be released the connection has previously been tightened with a too high oil pressure, possible resulting in overstretched studs. In this case increase the oil pressure till setting value + 10%. More than 10% is not allowed as other construction parts can be damaged.

If, at 10% overpressure, the nuts can be turned off always replace the studs as they may be overstretched.

If the nuts are still not free, even at an increased oil pressure, remove the tools, cut the nuts off and remove the studs. Fit new studs and nuts.

If, due to any cause, doubt exist regarding the stress in hydraulically tensioned studs the loosening value must be found.

- 4** Increase the oil pressure in the hydraulic jacks in steps e.g. of 50 bar.
- 5** Check after each step whether the nuts can be turned.
- 6** If this value more than 50 bar below the stated value is, action must be taken.

2.4.3.10. Tightening of hydraulically tightened connection

In order to organise the hydraulic procedure (jack pressure and tightening procedure) four steps are defined:

First step, positioning of engine components and pre-stress

Turn manually the nuts on the studs and check the nuts moves easily without feeling any restriction. Tighten the nuts with the tool pin mount the hydraulic tools and pressurize the hydraulic jacks to the setting value for the pre-stress.

This pressure is applied to position the engine components to be assembled. Turn the nuts on with the tool pin. Release slowly the oil pressure in the hydraulic jacks to prevent damage to the calibrated oil pressure gauge. Now we have obtained a basis for the tightening procedure.

Second step, setting of materials

Pressurize the jacks till setting value of the final stress. Turn the nuts with the tool pin and count the number of holes in the circumference of the nuts the nuts can be shifted.

When all nuts can be shifted the same number of holes we then have the indication that the elongation of all studs is the same and that is how it should be.

In case one or more nuts can't be turned the same distance, measured on the circumference on the nuts compared to the others, remove and inspect the tools on good functioning and check the engine components for correct position.

After releasing the jack pressure the force will set the engine components and nuts and the remaining stress in the studs becomes somewhat lower.

Third step, obtaining the correct prestress

To correct for "setting of materials" the jacks have to be pressurized again to the pressure of the value setting of the final stress.

Now the engine components are set it is possible to turn all nuts a little further. After the jack pressure is released the correct final stress in the studs will exist.

Fourth step, checking the correct final stress

To make absolutely sure the connection is correct we have to make a "checking step". Pressurize the jacks again till setting value and try, with the tool pin, to turn the nuts any further. This may **not** be possible. Check with a feeler blade of 0.03 mm that no clearance exists between nut and contact surface.

If that is the case the studs have the required prestress and the components correctly connected.

Warning!

In case it is possible to tighten the nuts more than the prescribed number of holes or there is still clearance between nut and contact surface, the connection is NOT safe. In such a case all components have to be disconnected and examined:

- on deviation in material
 - wrong assembling
 - wrong assembling procedure.
-

2.4.3.11. Torque spanner

Modern technology requires more and more the fastening of screwed connections every time at the same prestress within certain tolerances. Human forces exerted on spanners are not sufficient accurate to obtain the required result c.q. the fastening torque. To make this possible the application of a torque spanner becomes a must.

To apply a torque spanner the following must be known to avoid mistakes resulting in unsafe connections.

- A torque spanner is a precision instrument that should be handled with utmost care. Every time the torque spanner is used the spanner should be restored in its box.
- The spanner should be kept clean, not dropped and not drenched with water or dirty oil.
- The spanner is calibrated between 20% and 100% of its scale and has a tolerance within 4% of its reading. Take care the spanner is periodically recalibrated.
- Calibrate the torque spanner every time in the horizontal or vertical plane to compensate for the spanner weight.
- Never use the spanner over the 100% scale. Distortion may exist and readings may become inaccurate.

- Always apply force on the torque spanner in a clockwise rotation, the direction shown on the spanner and only apply the exerted force in the centre of the knurled handle. The spanner clicks and gives a few degrees of free movement when the desired torque is reached. When you completely release the force, the spanner automatically resets for the next torque application.
- Never use a torque spanner to release connections.
- A high degree of torque accuracy can be obtained in the application of the correct lubricant on the thread of the connection. Where the contrary is not mentioned only use engine lubricating oil.
- Support the tool well during tightening.
The tightening torques are torque spanner readings and should be realised by an even tightening of bolt and nut. (if applicable, crosswise).
- Always apply force on a torque spanner in a slow on-going move not passing the required setting. Using the spanner in a fast move will easily do pass the required setting.
- The ratchet requires periodic cleaning and lubrication to ensure proper operation. Lubricate with a light oil. **DO NOT USE GREASE.**

Torque spanner set 9612ZT230 consists of spanner (1), extension (2), ring spanner (4) and bit (5). The scale is calibrated for torque spanner with extension. For convenience, extension (2) may be used together with ratchet (3), see fig. 2.4 – 10 .

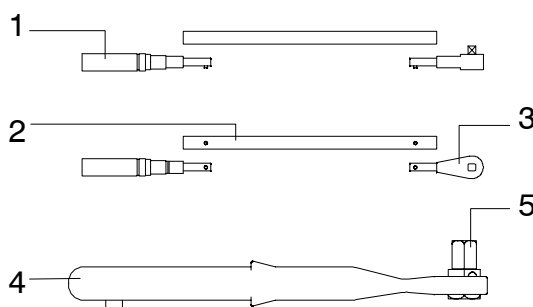


Fig. 2.4 – 10 Torque spanner

2.4.4. Tightening torque and jack pressure

2.4.4.1. Lubricating oil system components

(Chapter 1.2.)

Pos.	Connection	Torque Nm
1.	Lubricating oil pump	80
2.	Gearwheel on shaft	65

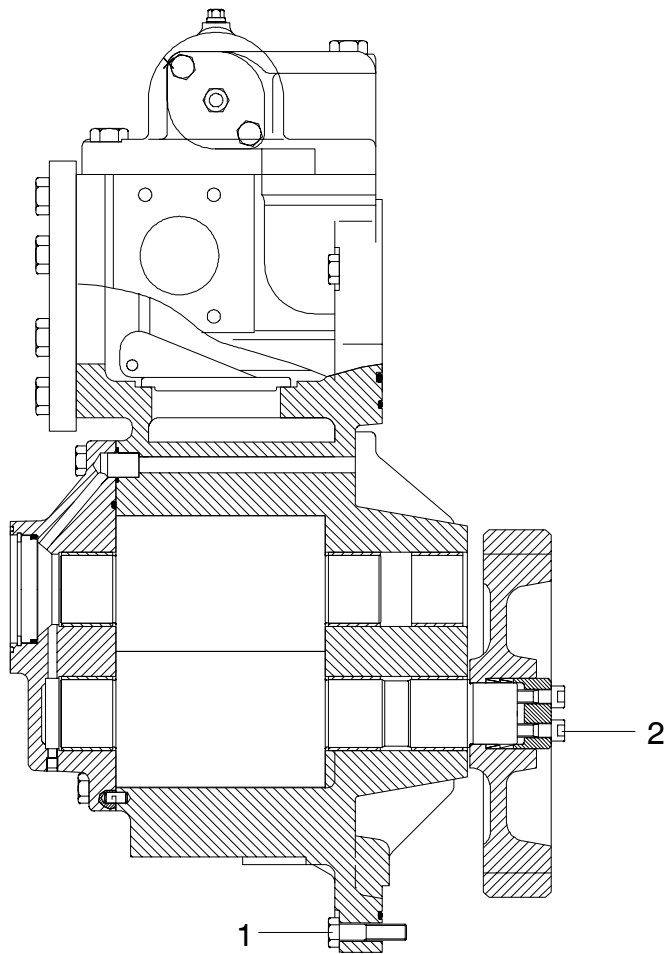


Fig. 2.4 – 17 Lubricating oil pump

2.4.4.2. Cooling water system components

(Chapter 1.4.)

Pos.	Connection	Torque Nm
1.	Cooling water pump	195
2.	Gear wheel on shaft	15.5
3.	Suction house	80
4.	Impeller on shaft	17

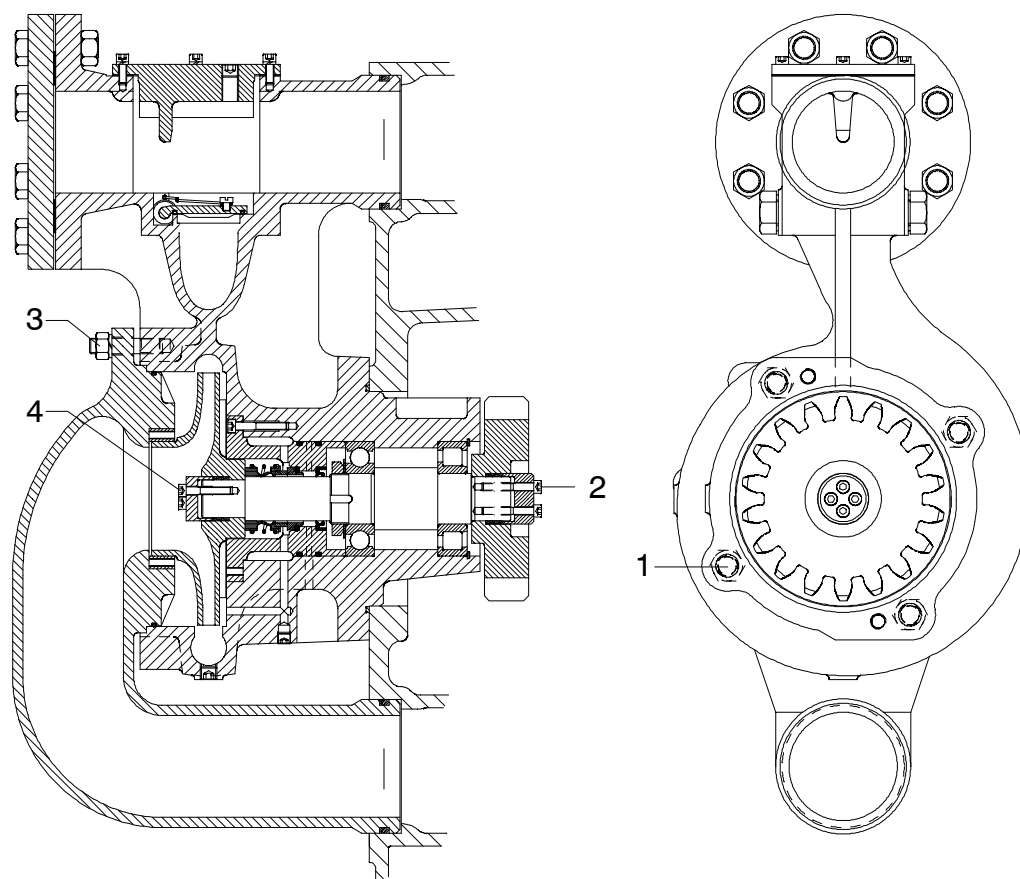


Fig. 2.4 – 18 Cooling water pump

2.4.4.3. Control system components

(Chapter 1.6.)

Pos.	Connection	Torque Nm
1.	Governor drive housing	175
2.	Gear wheel on shaft	10
3.	Governor	80
4.	Adapter	80

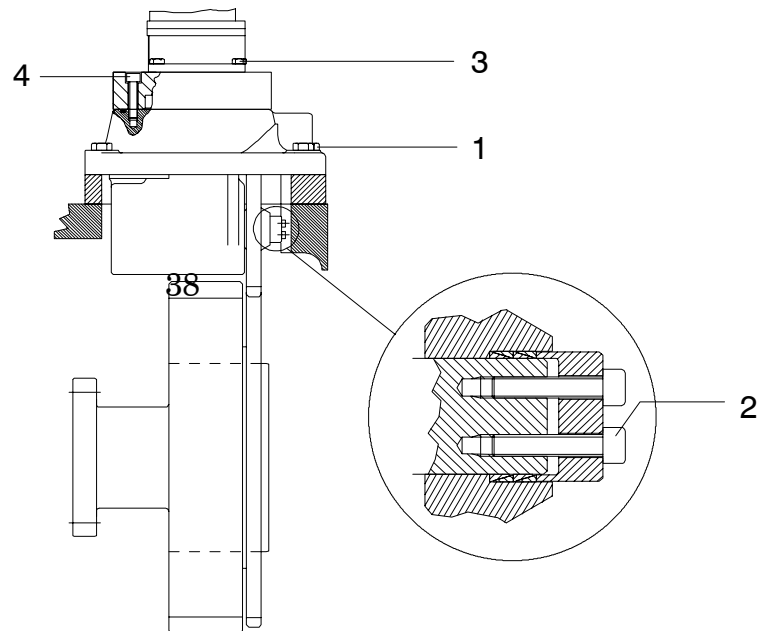


Fig. 2.4 – 19 Governor drive

2.4.4.4. Engine block with bearings

Pos.	Connection	Torque / Jack pressure	Elongation stud mm
1.	Main bearing stud , mounting	200 Nm	
2.	Main bearing cap, tightening first step second step third step Elongation	tool nr. 9612ZT903 200 bar 900 bar 900 bar	Measure size 'X' Measure size 'Y' 'Y'-'X' = 1.4–1.7
3+4	Side stud, mounting	0 Nm 360° back	
3.	Side stud (long), tightening first step second step third step Elongation	tool nr. 9612ZT902 100 bar 450 bar 450 bar	Measure size 'X' Measure size 'Y' 'Y'-'X' = 1.35–1.6
4.	Side stud (short), tightening first step second step third step Elongation	tool nr. 9612ZT902 100 bar 450 bar 450 bar	Measure size 'X' Measure size 'Y' 'Y'-'X' = 0.9 – 1.1

Note! The process of stud stretching is mentioned in chapter 2.5.2.4.

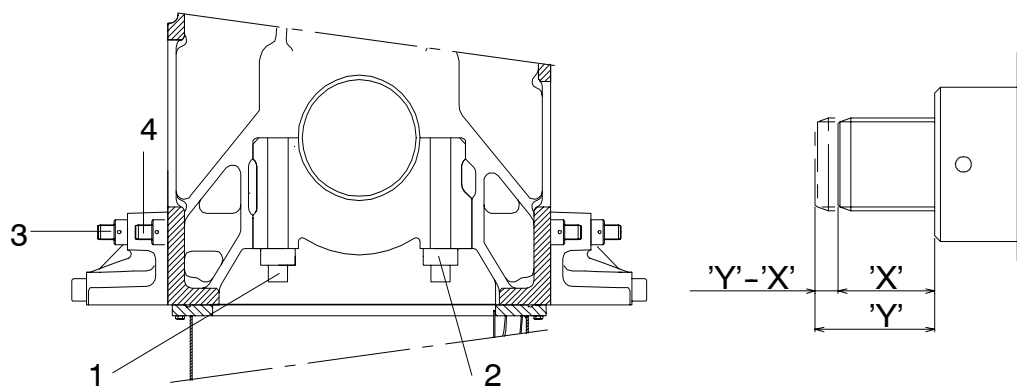


Fig. 2.4 – 20 Main bearing

2.4.4.5. Crankshaft, connecting rod, piston

(Chapter 2.6.)

Pos.	Connection	Torque Nm
1.	Vibration damper	2400
2.	Gear wheel	195
3.	Extension piece	195

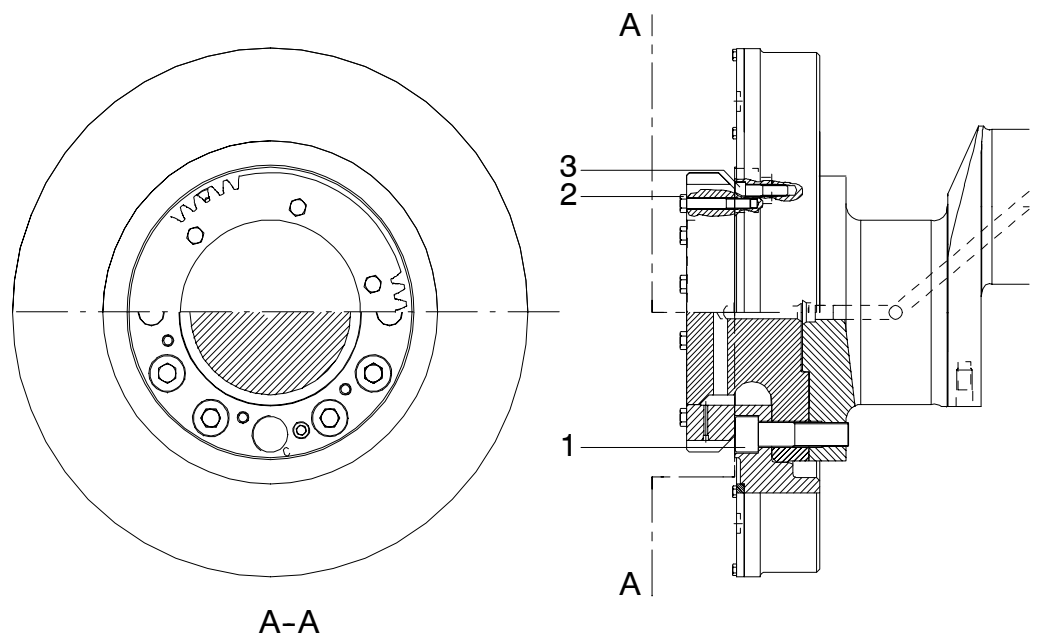


Fig. 2.4 – 21 Vibration damper

Pos.	Connection	Torque Nm
1.	Split gear wheel on crankshaft	270
2.	Split gear wheel on crankshaft	540

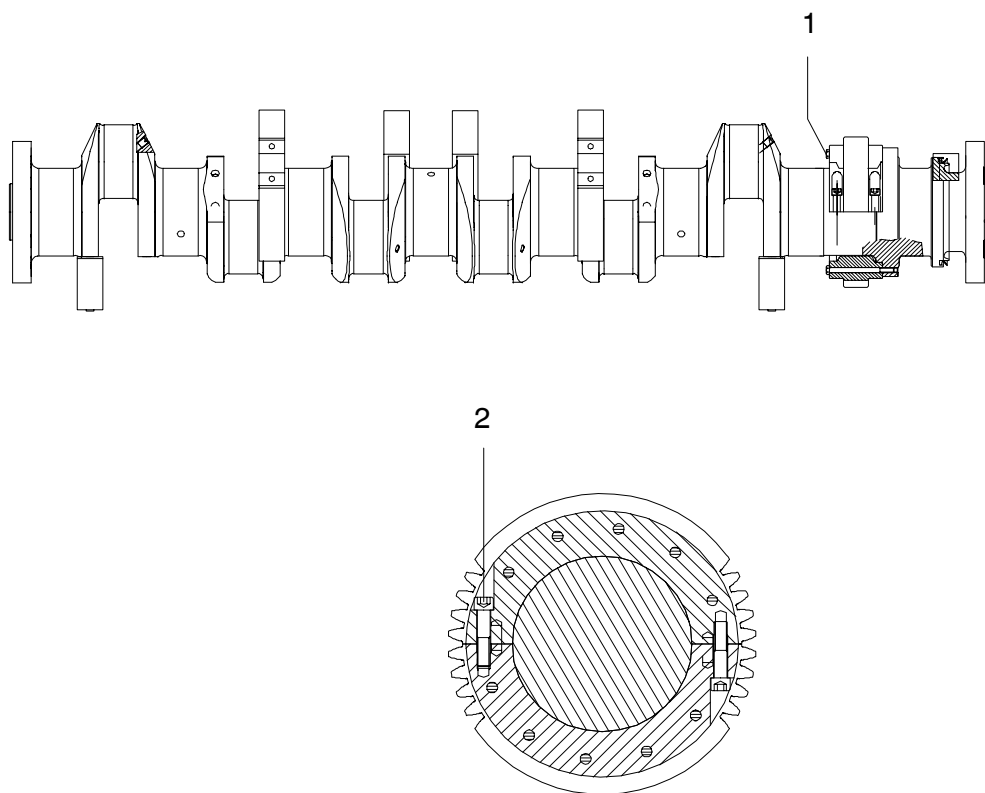
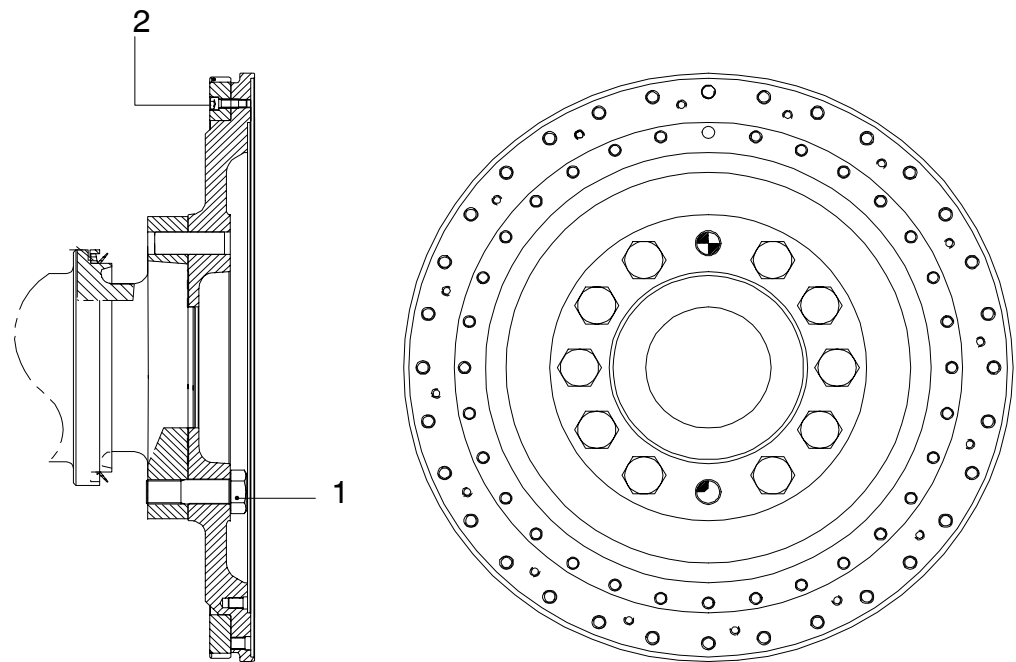


Fig. 2.4 – 22 Crankshaft gear

Pos.	Connection	Torque Nm
1.	Flywheel	2400
2.	Toothed ring	80

**Fig. 2.4 – 23 Flywheel**

Pos.	Connection	Torque / Jack pressure	Elongation stud mm
1.	Big end stud, mounting	100 Nm	
2.	Big end bearing cap, tightening	tool nr. 9612ZT944	
	first step	380 bar	Measure size 'X'
	second step	970 bar	
	third step	970 bar	Measure size 'Y'
	Elongation		'Y'-'X' = 0.7 – 0.9

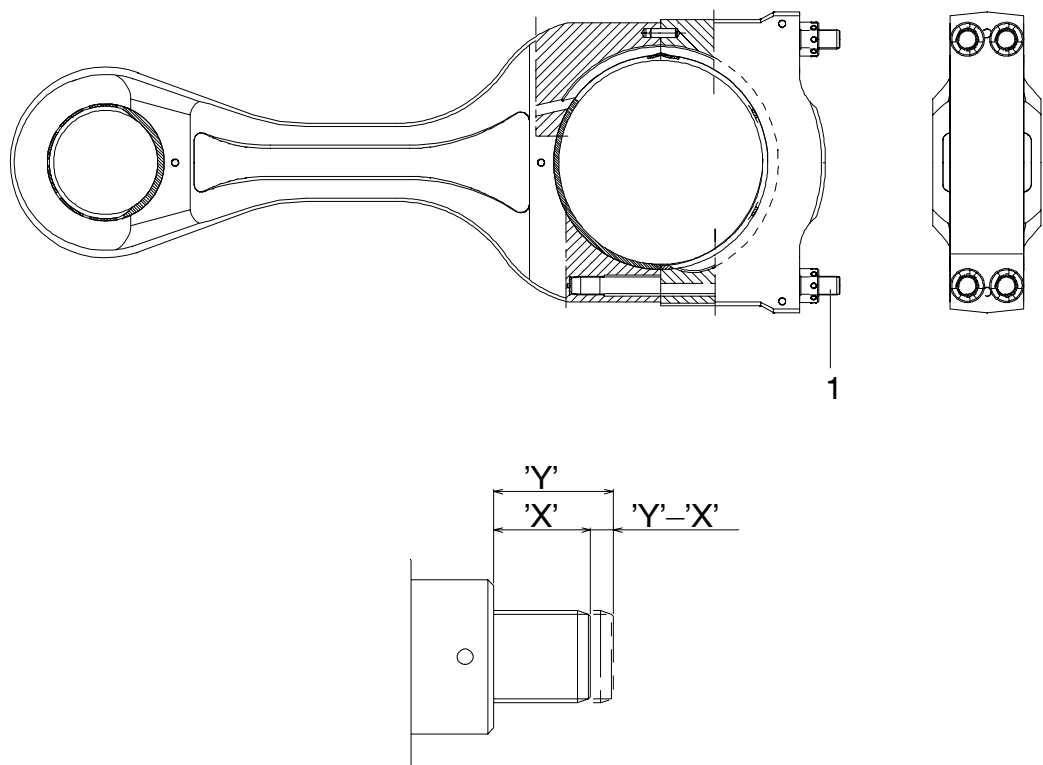


Fig. 2.4 – 24 Big end bearing

2.4.4.6. Cylinder head with valves

(Chapter 2.7.)

Pos.	Connection	Torque Nm
1.	Multiduct to cylinder head	200
2.	Multiduct to exhaust gas manifold	200
3.	Multiduct to engine block:	79

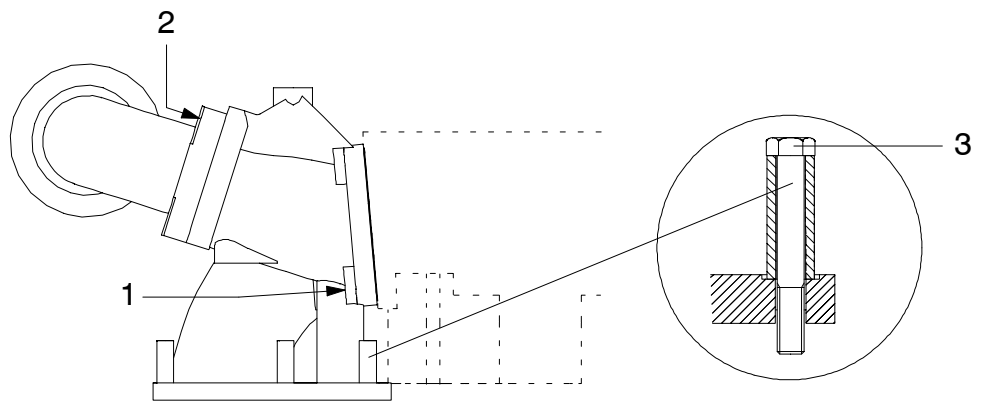


Fig. 2.4 – 25 Multiduct connection

Pos.	Connection	Torque Nm
1.	Roller holder casing	46

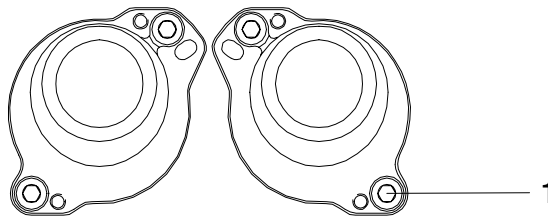


Fig. 2.4 – 26 Roller holder casing

Pos.	Connection	Torque / Jack pressure	Stud elongation mm
1.	Cylinder head stud, mounting	200 Nm	
1.	Cylinder head, tightening first step second step third step Elongation	tool nr. 9612ZT904 300 bar 725 bar 725 bar	Measure size 'X' Measure size 'Y' $'Y' - 'X' = 1.15 - 1.30$

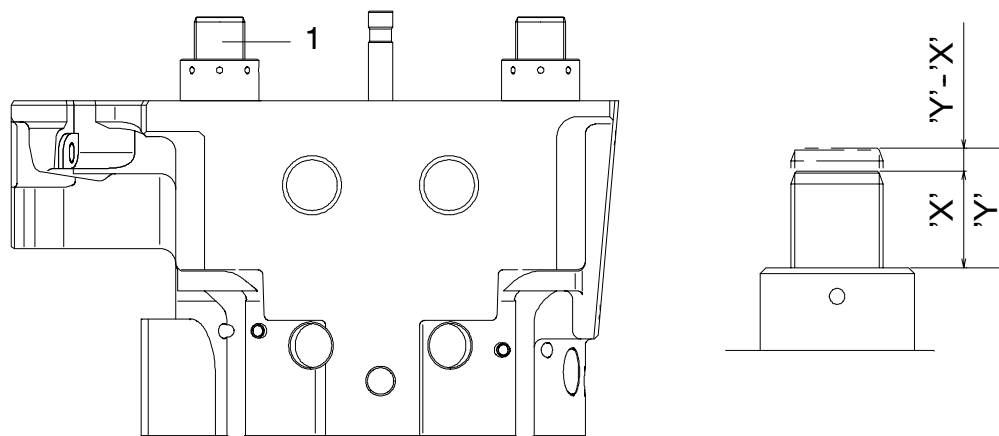


Fig. 2.4 – 27 Cylinder head

Pos.	Connection	Torque Nm
1.	Rocker arm bracket	385
2.	Locking nut	120
3.	Locking nut	220

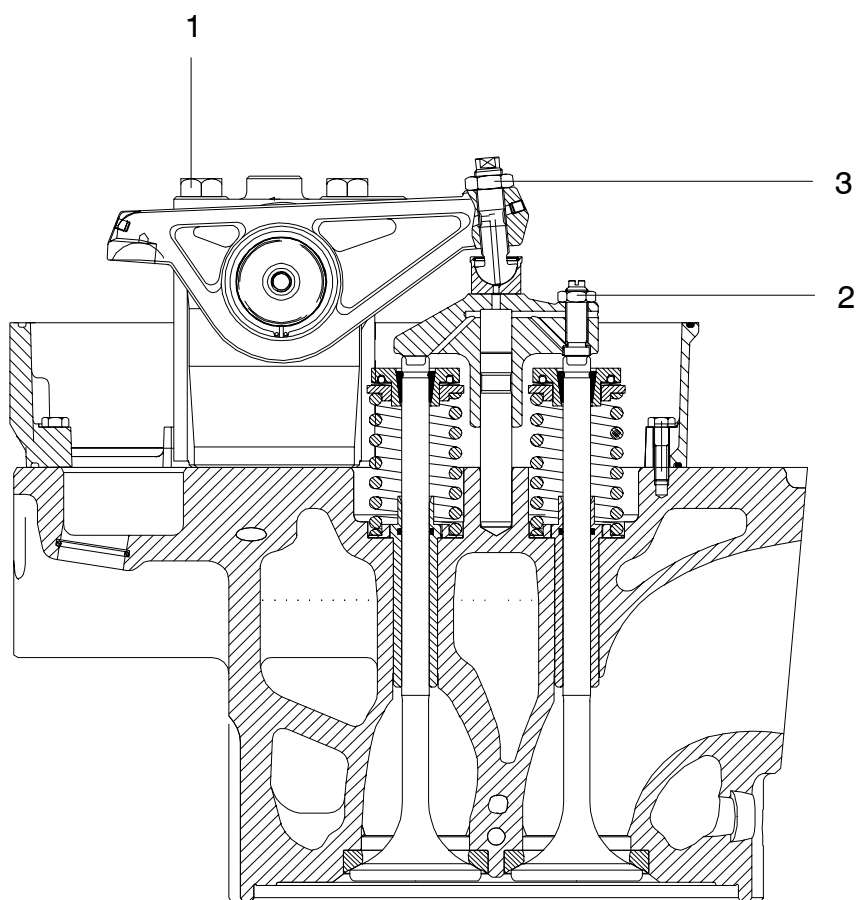


Fig. 2.4 – 28 Rocker arm mechanism

2.4.4.7. Camshaft and valve drive

(Chapter 2.8.)

Pos.	Connection	Torque Nm
1.	Intermediate gear wheel shaft cover	46
2.	Gear wheel	650

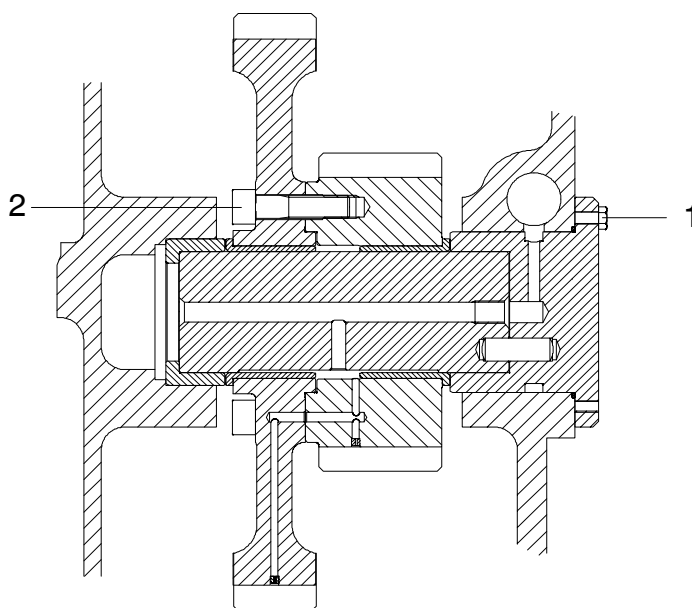


Fig. 2.4 – 29 Intermediate gear shaft

Pos.	Connection	Torque Nm
1.	Camshaft section to journal: first step second step (final setting)	40 330
2.	Thrust ring to journal	80
3.	Journal to camshaft gear wheel	195

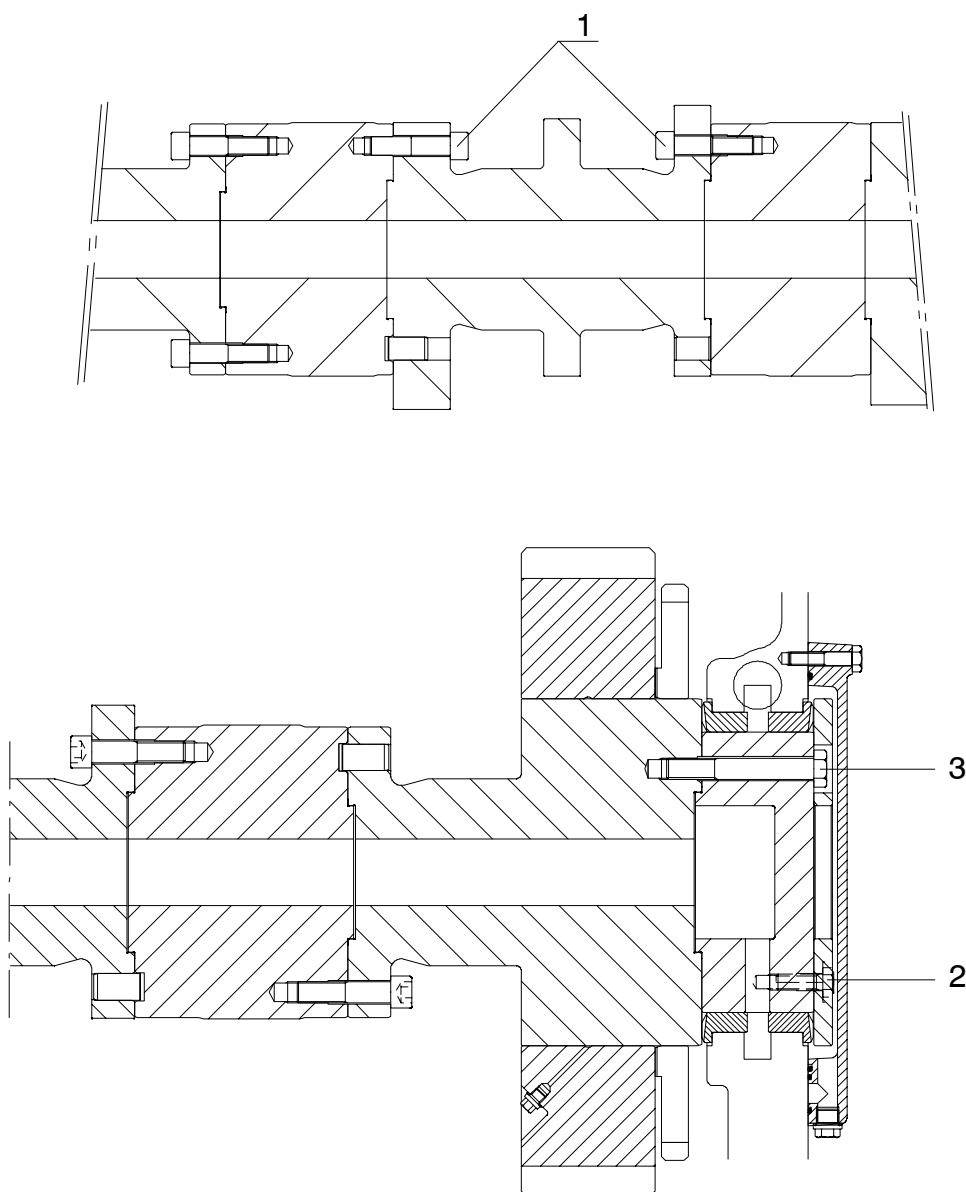


Fig. 2.4 – 30 Camshaft / camshaft gearwheel

Pos.	Connection	Torque Nm
1.	Drive shaft nut	100
2.	Locking bolt	10

Note! Make adjustments only with required tools.

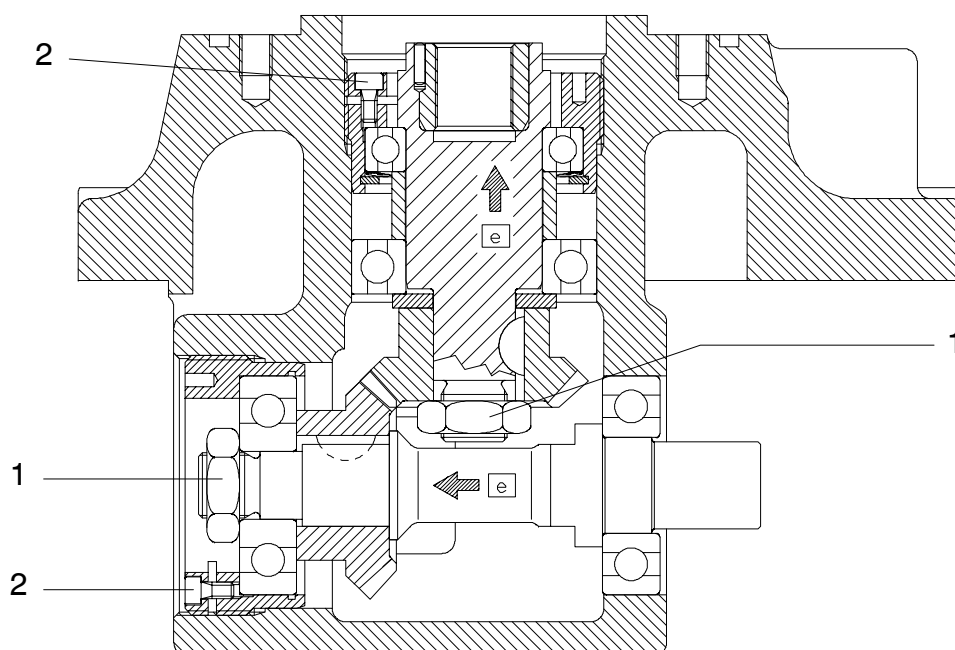


Fig. 2.4 – 31 Governor drive

2.4.4.8. Injection system components

(Chapter 2.9.)

Pos.	Connection	Torque Nm
1.	Fuel injector nut	30
2.	Stud	10

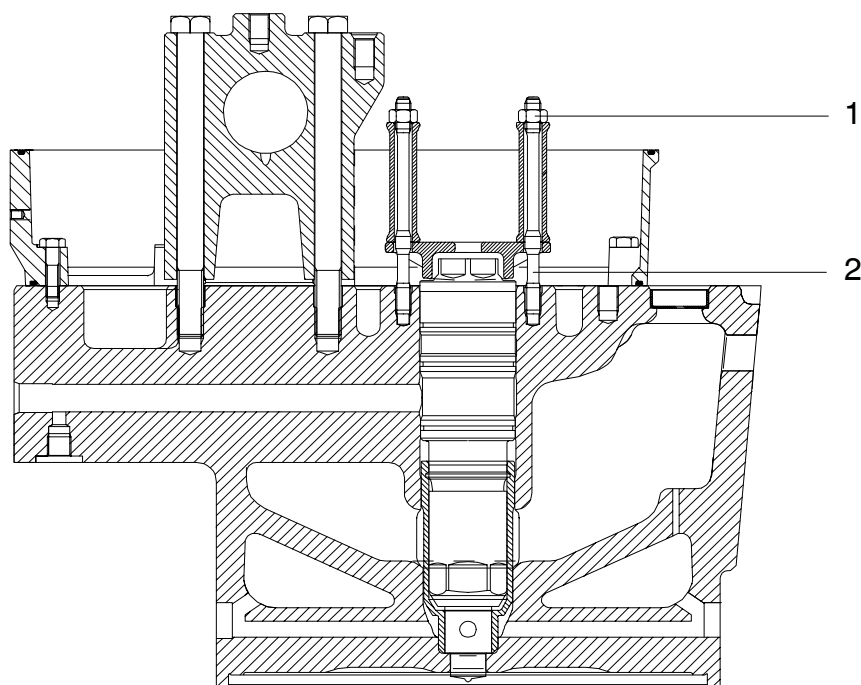


Fig. 2.4 – 32 Fuel injector mounted

Pos.	Connection	Torque Nm
1.	Connecting pipe	60
2.	HP fuel line union	60

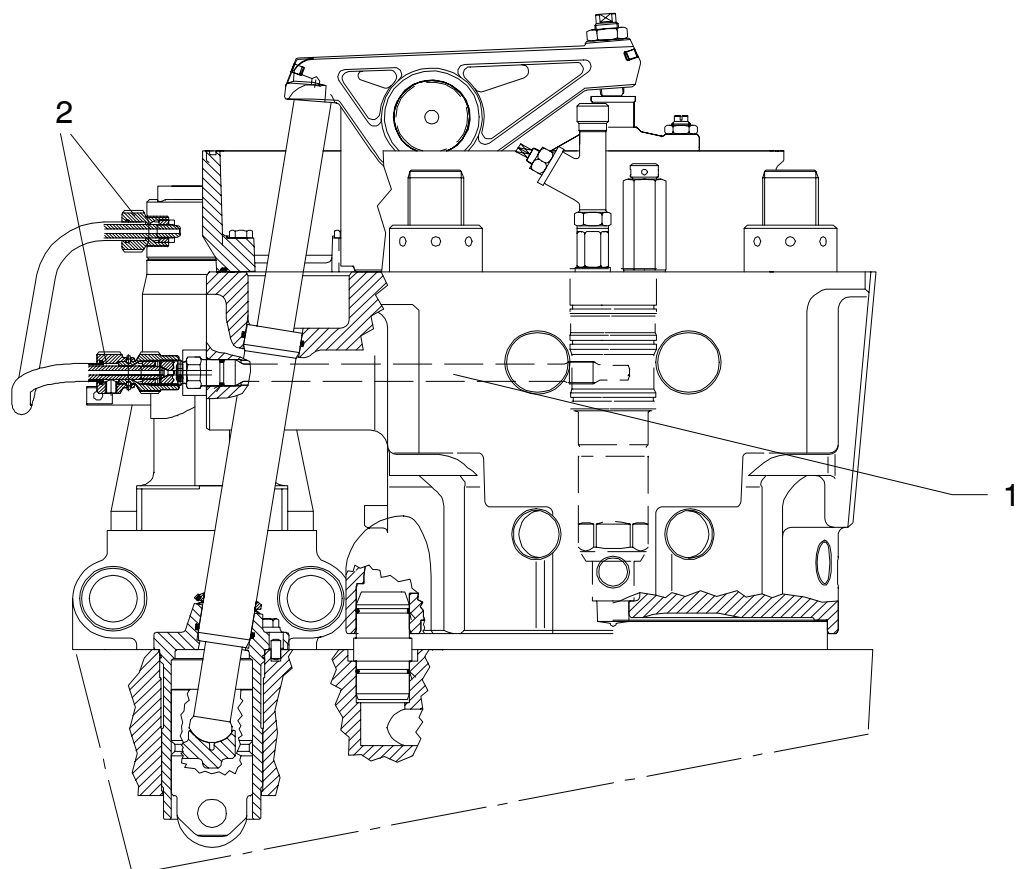


Fig. 2.4 – 33 Fuel injector, HP fuel line

Pos.	Connection	Torque Nm
1.	Nozzle (Use loctite 767 or equivalent on thread)	750..800
2.	Locking nut	70..90
3.	Plug	20..22

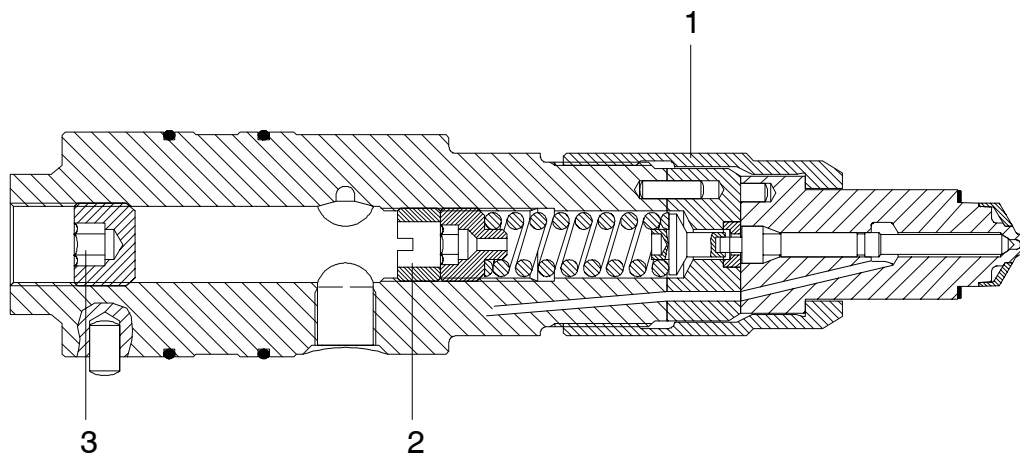


Fig. 2.4 – 34 Fuel injector

Pos.	Connection	Torque Nm
1.	HP fuel pump	140
2.	Fuel pump support	46

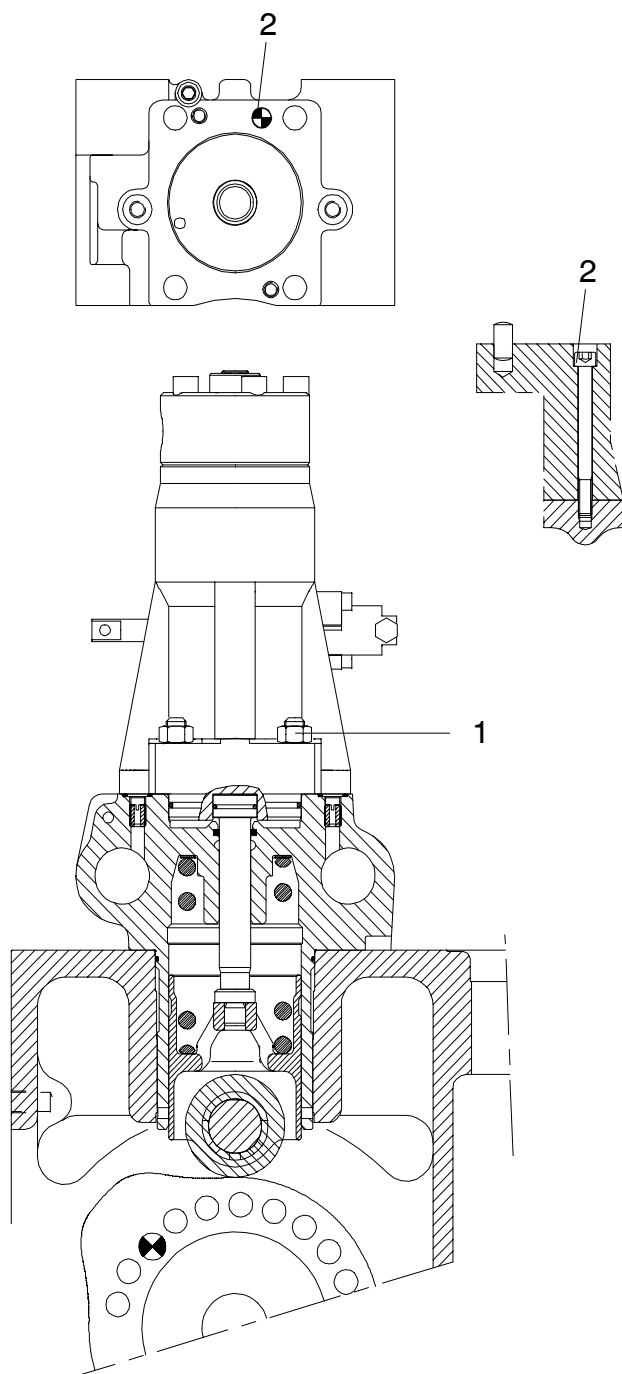


Fig. 2.4 – 35 HP fuel pump drive and tappet housing

2.4.4.9. General table of tightening torques for not specified bolt connections

When the tightening torque is not specified, use the values from the table below.

Unless stated otherwise all threads and contact surfaces of the nuts and bolts should be sparingly lubricated with engine oil before tightening.

For general information, see also:

- DIN 13
- "VDI Richtliniën 2230"
- DIN – 912 – 931 – 933 – 6912 – 7984.

Norm bolt connections			
Thread Dim.	Pitch	Bolt class	Torque Nm
M4	0,7	8.8	2,8
		10.9	4,1
		12.9	4,8
M5	0,8	8.8	5,5
		10.9	8,1
		12.9	9,5
M6	1,0	8.8	9,5
		10.9	14,0
		12.9	16,5
M8	1,25	8.8	23,0
		10.9	34,0
		12.9	40,0
M10	1,5	8.8	46,0
		10.9	68,0
		12.9	79,0
M12	1,75	8.8	79,0
		10.9	117,0

Norm bolt connections			
Thread Dim.	Pitch	Bolt class	Torque Nm
		12.9	135,0
		8.8	125,0
M14	2,0	10.9	185,0
		12.9	215,0
		8.8	195,0
M16	2,0	10.9	280,0
		12.9	330,0
		8.8	280,0
M18	2,5	10.9	390,0
		12.9	460,0
		8.8	390,0
M20	2,5	10.9	560,0
		12.9	650,0
		8.8	530,0
M22	2,5	10.9	750,0
		12.9	880,0
		8.8	670,0
M24	3,0	10.9	960,0
		12.9	1120,0
		8.8	1350,0
M30	3,5	10.9	1900,0
		12.9	2250,0

2.4.5. Adjustments and Tolerances

2.4.5.1. Adjustments

Timing, clearances and settings	
Valve clearances cold engine:	
– Inlet valve	0.7 mm
– Exhaust valve	0.8 mm
Fuel delivery commencement	See test records
Opening pressure fuel injector needle	600 bar
Nominal speed 900 / 1000 rpm	Overspeed stop 1035 / 1150 $\pm$ 10 rpm

2.4.5.2. Tolerances

	Part, measuring point
	Sleeve bearings The engine bearings consist of a steel shell onto which a very tiny bonding of almost pure Aluminium. On this bonding is a running layer of Aluminium alloy. It concerns bearings for: – crankshaft – connecting rod big end – camshaft A bearing is suitable for use as long as: – the shell thickness is expected to stay within the given tolerances, until the next inspection, (see this chapter). – the inside diameter is within the given tolerances, until the next inspection, (see this chapter). – the bearing shell is free of damage. – the bearing shell is free of corrosion. – the wear pattern is equal. – the running layer is not overloaded. An overloaded bearing shell can be recognized by locally melted or smeared lining. New bearings are treated with a corrosion protection that has to be removed before mounting. Where bearings show heavy wear grooves in the running layer, the quality of the lubricating oil cleaning should be observed more carefully.

Note! For determination of wear, engine components and measuring equipment should for some hours first be acclimated at room temperature (20 °C).

	Part, measuring point	Design measurements		Nominal clearance mm
		Max. [mm]	Min. [mm]	
1.6.	Governor			
	Backlash driving gear *			0.12-0.18

* Shaft clearance in direction "e" is zero.

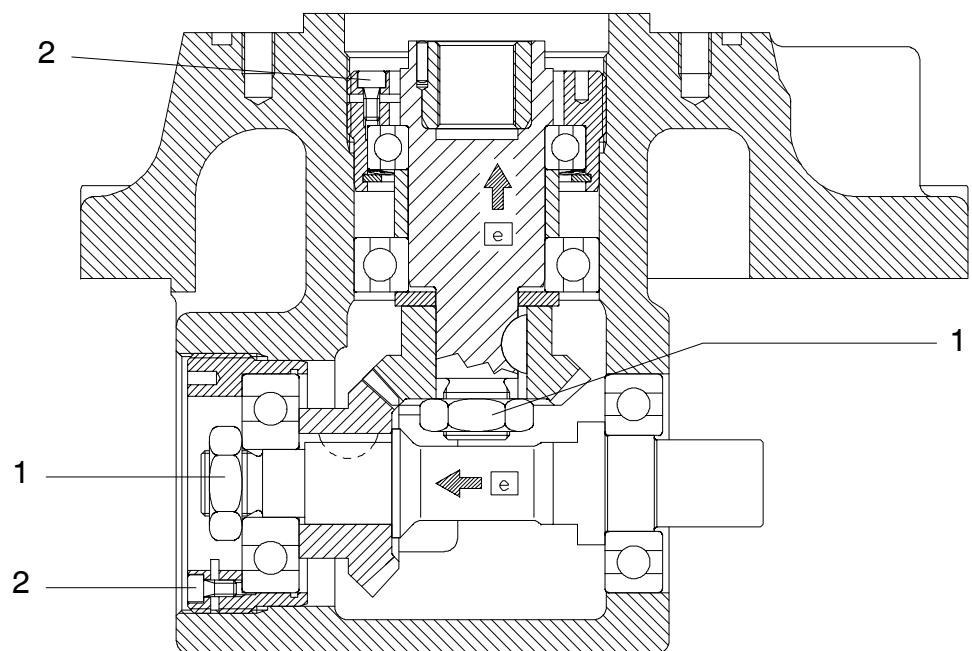


Fig. 2.4 – 30 Governor drive

	Part, measuring point	Design measurements		Nominal clearance mm
		Max. [mm]	Min. [mm]	
1.2.	Lubricating oil pump			
	Backlash driving gear wheel (1) lubricating oil pump gear wheel (2)			0.49-0.77
	Cooling water pump (chapter 1.4.)			
	Backlash driving gear wheel (1) H.T. cooling water pump gear wheel (3)			0.57-0.87
	Backlash driving gear wheel (1) L.T. cooling water pump gear wheel (4)			0.55-0.80

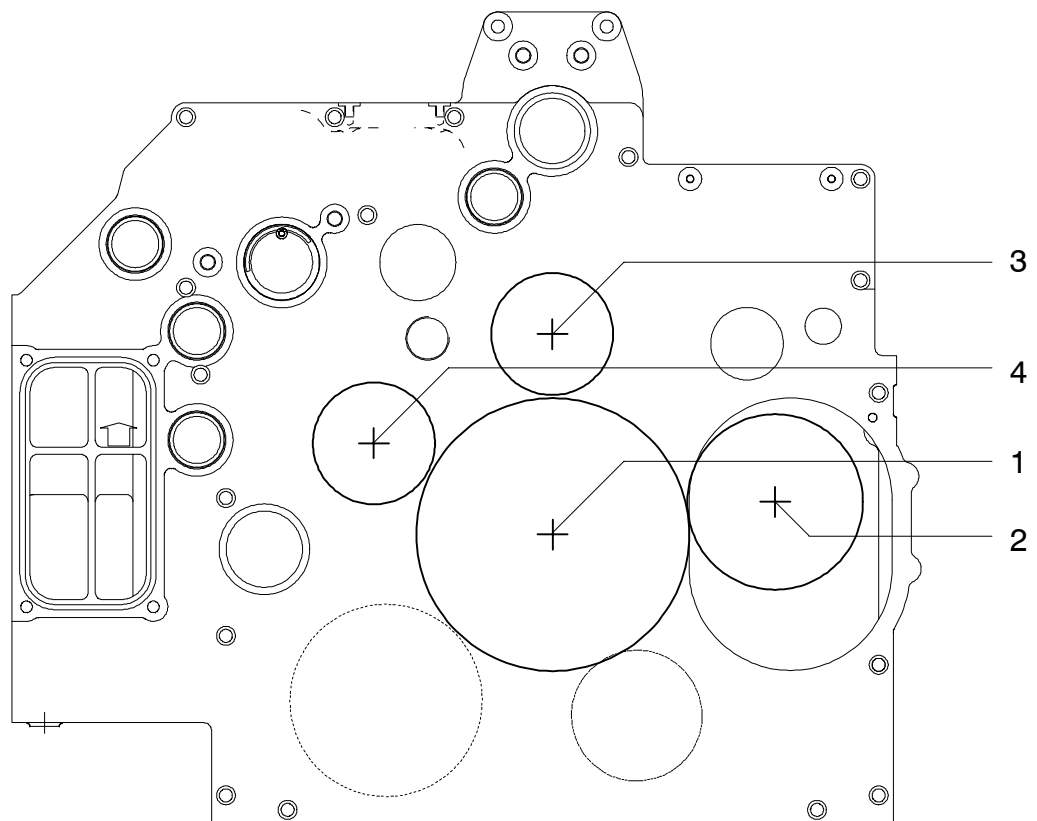
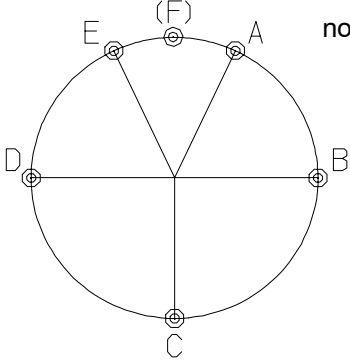


Fig. 2.4 – 31 Gear wheel train

	Part, measuring point	Design measurements		Nominal clearance mm
		Max. [mm]	Min. [mm]	
2.5.	Main bearing			
	Main bearing shell thickness	9.885	9.865	
	Diameter of main bearing housing	300.032	300.000	
	Assembled bearing diameter	280.329	280.258	
	Main bearing clearance (flywheel bearing)			0.258–0.361
	Thrust bearing axial clearance			0.15 – 0.40
	Thrust bearing thickness	11.000	10.950	
2.5.	Camshaft bearing			
	Camshaft diameter	220.000	219.971	
	Camshaft bearing bush thickness	9.910	9.890	
	Camshaft bearing housing bore	240.029	240.000	
	Camshaft thrust bearing bush thickness	14.908	14.879	
	Assembled bearing bore	220.274	220.205	
	Camshaft bearing clearance			0.205–0.303
	Camshaft thrust bearing housing bore	240.029	240.000	
	Camshaft thrust bearing diameter	210.000	209.971	
	Assembled thrust bearing bore	210.285	210.205	
	Camshaft thrust bearing clearance			0.205–0.314
	Camshaft thrust bearing axial clearance			0.2 – 0.5
	Camshaft thrust bearing axial thickness	7.90	7.85	
2.6.	Crankshaft			
	Crankshaft journal diameter	280.000	279.968	
	Crankshaft journal ovality	0.025	—	
	Crankshaft journal taper	0.015	—	
2.6.	Cylinder liner			
	Cylinder liner diameter	260.052	260.000	
	Cylinder liner ovality at TDC	0.02	—	
	No-go is 0.6 mm diameter oversize			

	Part, measuring point	Max. mm
2.6.	Crankshaft deflection criteria	
	Measuring the deflections of the crankshaft the commissioning report is considered as a reference. For in-line engines measuring place of dial gage on counter weight: 115 mm from mounting face crankshaft - counter weight The crankshaft deflection criteria: - free and driving end not coupled to the installation - all cylinders - driving end coupled to the installation - first cylinder - free and driving end coupled to the installation - first and last cylinder The deviation between adjacent cranks: - compare crankshaft deflections at the same measuring points (A-A; B-B etc.) (0.7 x crankshaft deflection limit (0.07)) For in-line engines the deviation on the same crank between A and E is : Checking the measurement results Average between A and E = $0.5(A+E) = F$ - free and driving end not coupled to the installation, for all cylinders: Deviation between F and B / B and C / C and D / D and F - driving end coupled to the installation, for first cylinder: Deviation between F and B / B and C / C and D / D and F - free and driving end coupled to the installation, for first and last cylinder: Deviation between F and B / B and C / C and D / D and F operating side (F) non-operating side  Note: - Mentioned values are with engine and foundation at ambient temperature. - If the values are > max.: - check measurement tools and measurement procedure, see section 2.6.1. - check foundation and alignment of the engine and driven shaft. Re-align if necessary.	0.07 0.16 0.16 0.05 0.015 0.035 0.080 0.080

	Part, measuring point	Design measurements		Nominal clearance mm
		Max. [mm]	Min. [mm]	
2.6.	Big end bearing			
	Big end bearing clearance			0.283–0.186
	Crank pin diameter	235.000	234.971	
	Crank pin ovality	0.020	—	
	Crank pin taper	0.015	—	
	Big end bearing shell thickness	5.914	5.895	
	Big end bore diameter	247.029	247.000	
	Big end bore ovality	0.015	—	
	Assembled bearing bore	235.254	235.186	
2.6.	Piston			
	Gudgeon pin diameter	120.000	119.985	
	Small end bore	135.025	135.000	
	Gudgeon pin bearing clearance			0.085–0.165
	Axial clearance small end bearing			0.4 – 0.7
	Small end bearing bush thickness	7.465	7.445	
	Clearance gudgeon pin – piston			0.045–0.085
	Bore diameter in piston	120.07	120.045	
	Piston – Compression ring 1 gap – Compression ring 2 gap – Oil scraper ring gap			0.6–0.9 1.8–2.1 0.8–1.15
	Piston ring axial clearance: – Compression ring 1 – Compression ring 2 – Oil scraper ring			0.223–0.265 0.173–0.215 0.053–0.095
	Piston ring groove height: – Groove 1 – Groove 2 – Groove 3	7.23 7.18 7.06	7.21 7.16 7.04	
	Piston clearance at bottom in cross direction of engine			0.134–0.228
	Corresponding piston diameter	259.866	259.824	

	Part, measuring point	Design measurements		Nominal clearance mm
		Max. [mm]	Min. [mm]	
2.7.	Cylinder head			
	Valve guide inner diameter	20.16	20.13	
	Valve stem diameter	20.000	19.979	
	Clearance valve stem / valve guide			0.130–0.181
	Inlet valve guide recess in cylinder head	105.03	105.00	
	Exhaust valve guide recess in cylinder head:			
	– outer bore	97.030	97.000	
– inner bore	79.430	79.400		

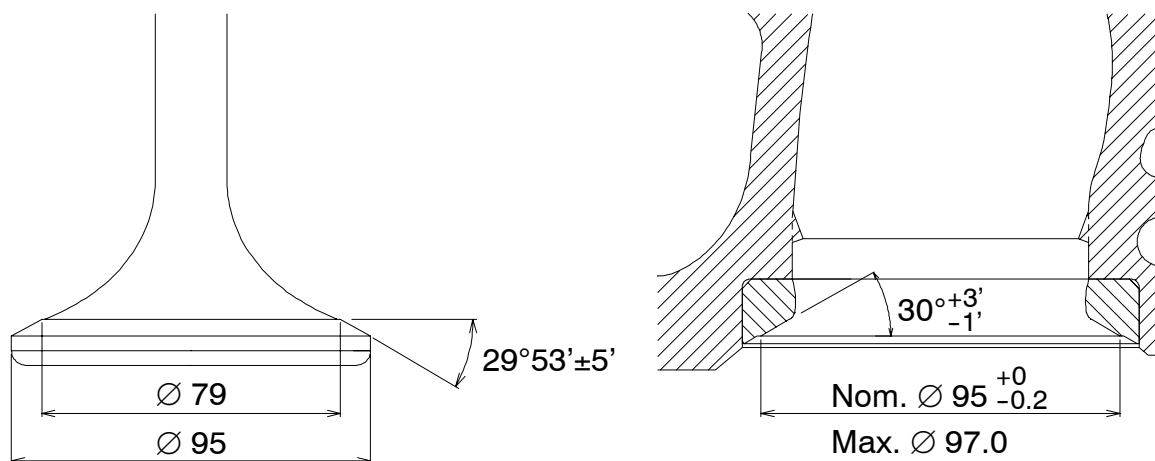


Fig. 2.4 – 32 Inlet valve and valve seat in cylinder head

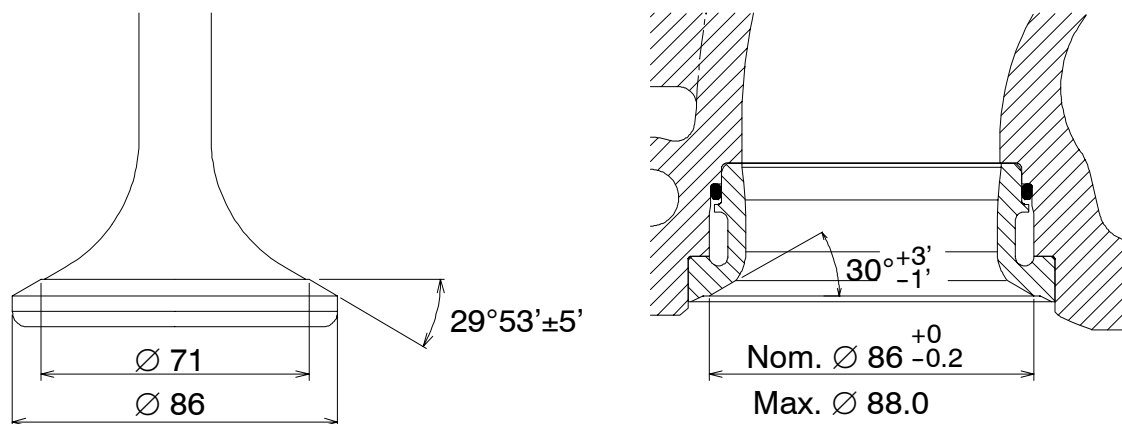


Fig. 2.4 – 33 Exhaust valve and valve seat in cylinder head

	Part, measuring point	Design measurements		Nominal clearance mm
		Max. [mm]	Min. [mm]	
2.8.	Intermediate gear wheels			
	Backlash crankshaft gear wheel (1) intermediate gear wheel (large) (2)			0.10-0.43
	Backlash intermediate gear wheel (3) camshaft gear wheel (4)			0.10-0.29
	Backlash gear wheel governor drive (5) gear wheel governor drive unit (6)			0.28-0.40
	Axial clearance intermediate gear wheel			0.30-1.50

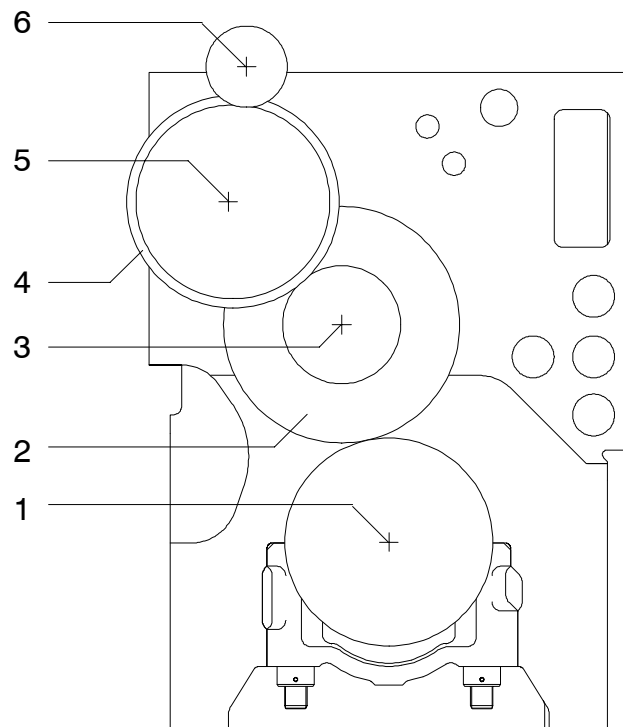


Fig. 2.4 – 34 Intermediate gear wheels

	Part, measuring point	Design measurements		Nominal clearance mm
		Max. [mm]	Min. [mm]	
2.8.	Valve drive			
	Roller holder casing inner diameter (1)	70.030	70.000	
	Roller holder (2)	69.940	69.910	
	Clearance roller holder / casing			0.060–0.120
	Pin bore in roller holder	28.021	28.000	
	Pin diameter (3)	28.000	27.987	
	Clearance pin / roller holder			0.000–0.034
	Bearing bush inner diameter (4)	28.050	28.030	
	Clearance pin / bearing bush			0.030–0.063
	Bearing bush outer diameter (4)	35.950	35.927	
	Roller inner diameter (5)	36.016	36.000	
	Clearance bearing bush / roller			0.050–0.089
	Rocker arm shaft diameter (6)	65.000	64.981	
	Shaft bore in rocker arm (7)	65.075	65.030	
	Clearance shaft / rocker arm			0.030–0.094
	Bridge piece pin diameter (8)	23.000	22.987	
	Pin bore in bridge piece (9)	23.065	23.024	
	Clearance pin / bridge piece			0.024–0.078

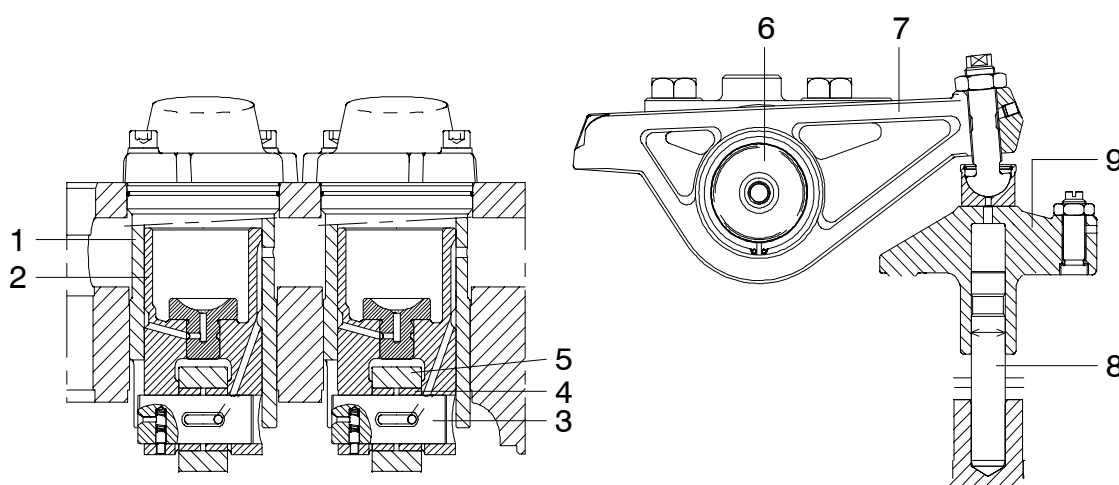


Fig. 2.4 – 35 Valve drive

	Part, measuring point	Design measurement		Nominal clearance mm
		Max. [mm]	Min. [mm]	
2.9.	Injection system			
	Nozzle needle lift 'A'	0.53	0.48	
	Distance 'B' fuel pump push rod	1.1	0.9	

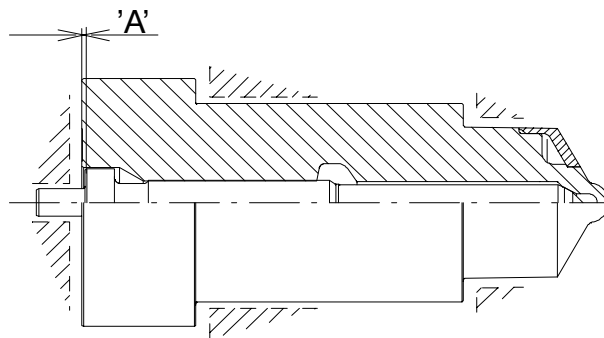


Fig. 2.4 – 36 Nozzle

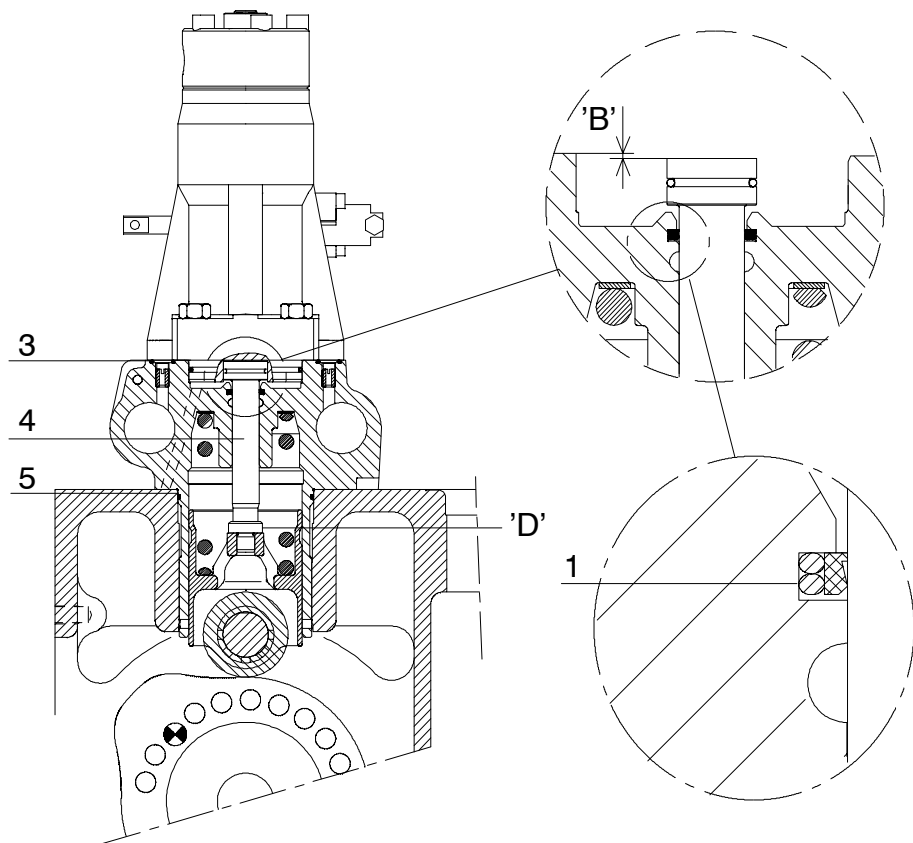


Fig. 2.4 – 37 HP fuel pump adjustment

2.4.6. Dimensions and masses

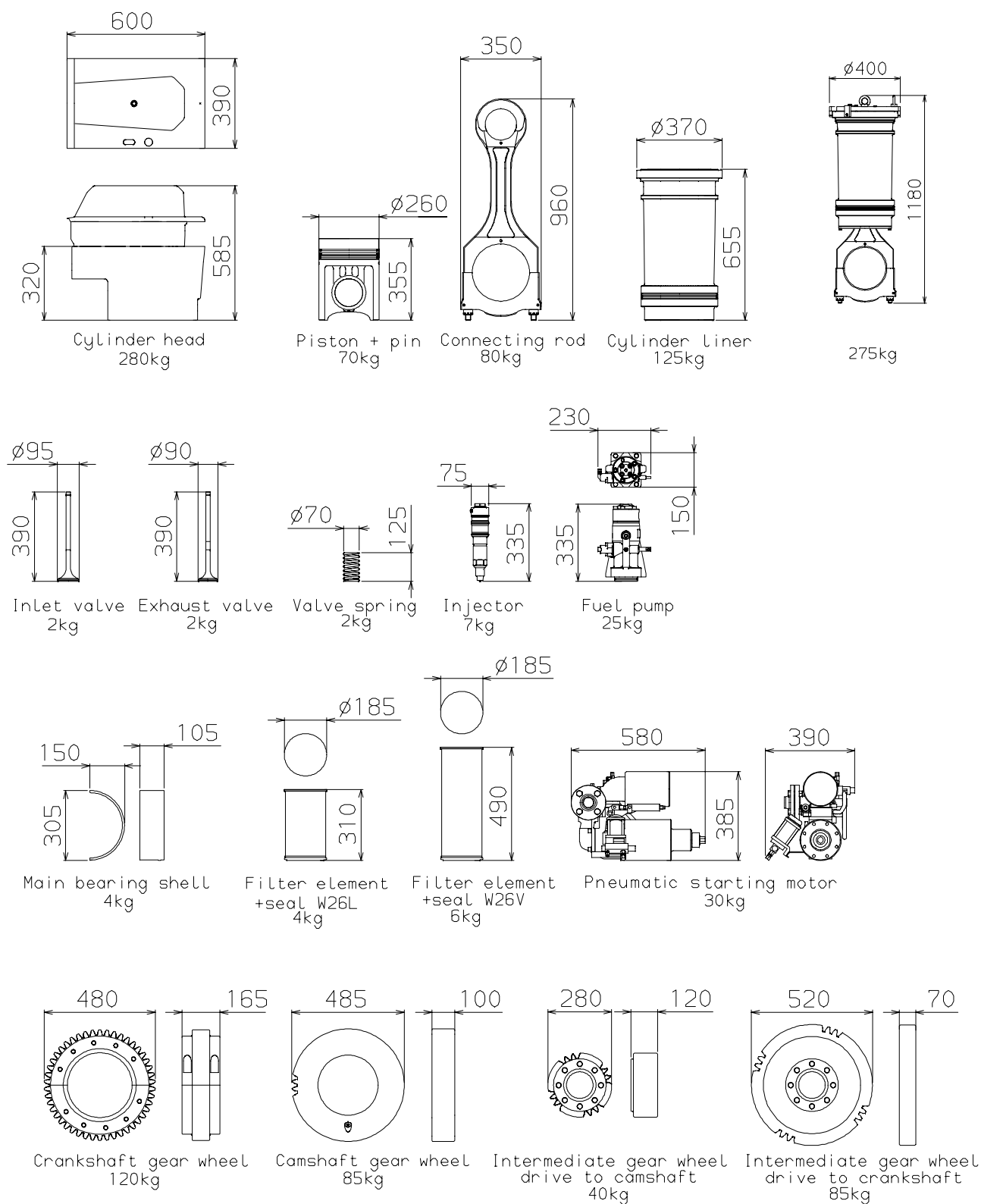


Fig. 2.4 – 38 Engine components

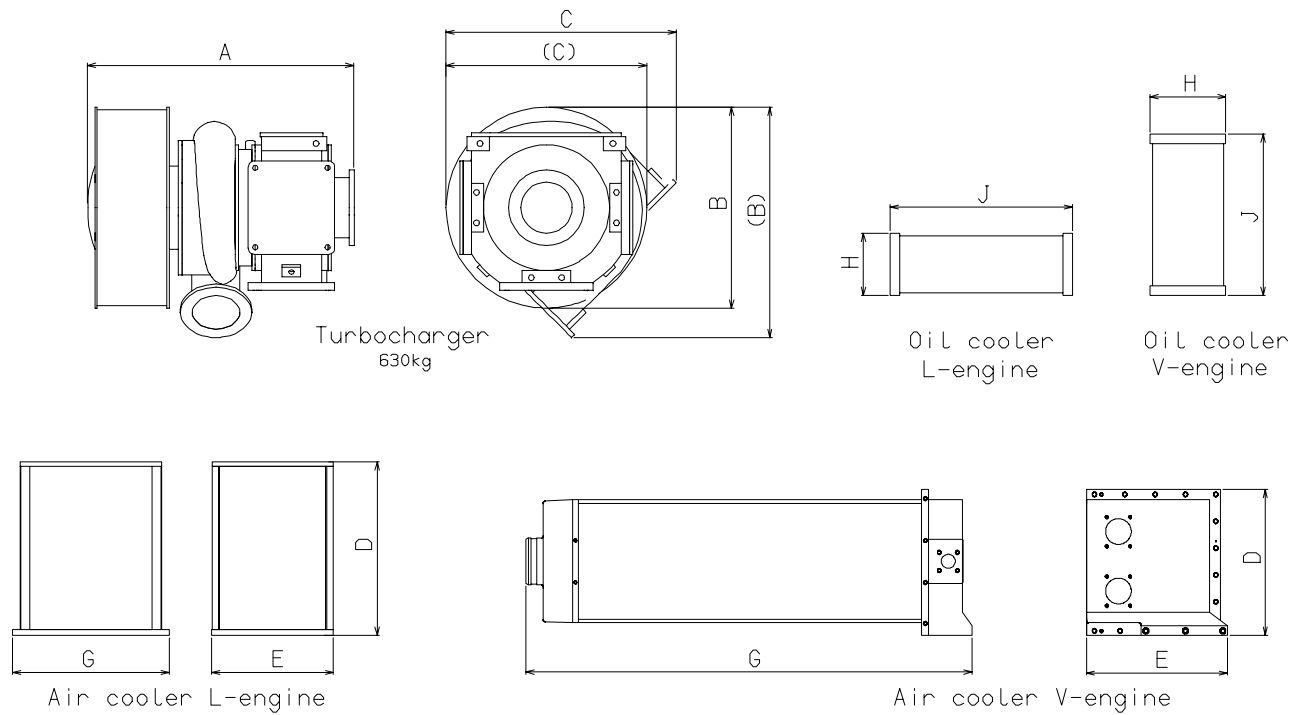


Fig. 2.4 – 39 Main parts

	6L26	8L26	9L26	12V26	16V26	18V26
A mm	1140	1140	1140	1140	1140	1140
B (B) mm	860 (990)	860 (990)	860 (990)	860	860	860
C (C) mm	900 (860)	900 (860)	900 (860)	990	990	990
D mm	745	745	745	625	625	625
E mm	525	525	525	605	605	605
G mm	673	673	673	1896	1896	1896
H mm	ø265	ø265	ø265	ø325	ø325	ø325
J mm	780	780	780	690	690	690
Oil cooler kg	40	40	40	2 x 60	2 x 60	2 x 60
Charge air cooler kg	285	375	375	680	725	725

